

Fixed Points and Short Cycles of Luce Permutations: Poisson Limits and Power-Law Endpoint Singularities

Abstract

The Luce, or Plackett–Luce, model generates a nonuniform random permutation by sampling without replacement with probabilities proportional to fixed weights. Its global asymptotic geometry was determined in [BCD25], where a Poisson law for the fixed points was conjectured. We prove this conjecture under weak multiscale profile and endpoint conditions. The fixed points converge to an inhomogeneous Poisson process, and the counts of cycles of any fixed collection of lengths converge jointly to independent Poisson random variables. Power-law endpoint singularities produce a second regime in which the cycle means diverge. We prove the corresponding joint central limit theorems and locate the cycles on logarithmic distance scales. When both endpoints contribute, the cycle counts arising from them are asymptotically independent and their means add.

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1 Introduction

Among nonuniform laws on permutations, the Luce model is one of the simplest to define. Give each label a positive weight and repeatedly choose from the remaining labels with probability proportional to weight. The rule is the same at every step and is consistent under deletion of the labels already chosen. Equivalently, attach independent exponential clocks to the labels and order their ringing times. The first description makes Luce’s law a natural model of sequential choice; the second makes it a tractable random permutation. It is therefore a natural setting in which to ask how a deterministic inhomogeneity changes the familiar microscopic statistics of a uniform permutation.

The asymptotic study of this object was initiated in [BCD25]. Starting from a macroscopic weight profile, that work identified the limiting permuton, computed pattern densities, proved local limits for consecutive relative patterns, and established a central limit theorem for inversions. These results give a detailed picture of the global geometry and of the relative order seen from a typical location. The same work then asked about a statistic that neither description sees: the number of fixed points. It predicted a Poisson law whose mean is the integral of the Luce permuton density along the diagonal [BCD25, Section 5.2]. That question is the starting point of this paper.

The obstacle is not identifying the candidate mean; it is reaching the scale on which a fixed point lives. A fixed point is the exact equality of a label and its rank. It lies on a lattice-scale diagonal of zero permuton area, whereas a consecutive-pattern limit records relative order rather than exact alignment. Permutations with the same permuton limit can consequently have completely different fixed-point behavior. A general cycle criterion in [Muk16] reaches the correct scale by assuming a uniform finite-dimensional local law, but the known Luce limits do not imply that hypothesis. The gap is most pronounced in the final positions, where few clocks remain and the limiting density may be singular.

Our first result proves the prediction in [BCD25] under minimal convergence assumptions on the weight profile and a multiscale condition at the final labels. We prove the spatial statement: the fixed points themselves converge to an inhomogeneous Poisson process, with intensity given by the diagonal of the Luce permuton density.

A fixed point is a cycle of length one, so the natural next question is whether the Poisson behavior persists for every fixed cycle length. It does. The counts of cycles of lengths $1, \dots, L$ converge jointly to independent Poisson variables, with means given by cyclic traces of the permuton density. This extension is not a formal product of one-point estimates: a cycle is a closed chain of exact assignments, and its vertices remain coupled through the common ranking.

Certain structured failures of endpoint tightness reveal a second regime. Very small weights ring late, while very large weights ring early. If a power-law singularity aligns one of these effects with the corresponding endpoint labels, then every logarithmic distance scale contributes a constant number of cycles. The means are then of order $\log n$. We prove the corresponding joint central limit theorems, identify the constants, and locate the cycles on logarithmic distance scales. The cycle counts near the left and right endpoints are asymptotically independent, and their contributions add in the total count. The size of the cycle intensity therefore separates two regimes: bounded intensities give Poisson limits, while diverging intensities give the corresponding centered and normalized Gaussian limits. The Sukhatme profile considered in [BCD25] already belongs to the second regime for fixed points.

The proof changes character at exactly the places suggested by this description. In the bulk, concentration of the exponential race identifies the predictable probability of an exact match.

At the last few positions, the remaining total weight may be too small to concentrate relatively; a counting argument avoids that denominator altogether. For longer cycles we resample a finite path of clocks into one background permutation, preserving the combinatorial constraint that each rank has one preimage. Finally, at a power-law endpoint we study the surviving population at large times or the arrived population at small times. Logarithmic distance turns the resulting ratio kernel into a translation-invariant one, and the cycle constants become its return densities.

Let $\theta_{n,1}, \dots, \theta_{n,n} > 0$. A Luce permutation is obtained by drawing the labels sequentially without replacement, with probability proportional to their weights. Thus, if $\pi_n(r)$ denotes the label drawn at step r , then

$$\mathbb{P}(\pi_n = \pi) = \prod_{r=1}^n \frac{\theta_{n,\pi(r)}}{\sum_{j=r}^n \theta_{n,\pi(j)}}. \quad (1)$$

We use $R_{n,i} = \pi_n^{-1}(i)$ for the rank of label i . The fixed-point count and process are

$$X_n := \sum_{k=1}^n I_{n,k}, \quad I_{n,k} := \mathbf{1}\{R_{n,k} = k\}, \quad \Xi_n := \sum_{k=1}^n I_{n,k} \delta_{k/n}. \quad (2)$$

This notation separates the draw order π_n from the rank map R_n ; the two are inverse permutations, while their fixed points agree.

The exponential-race representation is fundamental. Let $E_{n,i} \sim \text{Exp}(\theta_{n,i})$ be independent. Ordering these variables increasingly gives the Luce draw order, and

$$R_{n,i} = 1 + \sum_{j \neq i} \mathbf{1}\{E_{n,j} < E_{n,i}\}. \quad (3)$$

For $\ell \geq 1$, let

$$C_{n,\ell} := \frac{1}{\ell} \sum_{(i_1, \dots, i_\ell) \in [n]_\neq^\ell} \prod_{a=1}^{\ell} \mathbf{1}\{R_{n,i_a} = i_{a+1}\}, \quad i_{\ell+1} := i_1. \quad (4)$$

Thus $C_{n,1} = X_n$. The draw order and rank map are inverses, so they have the same cycle counts.

1.1 Context and relation to earlier work

The model originates in the theory of stochastic choice. The proportional-choice axiom was introduced in [Luc59], and the induced law on complete rankings was developed in [Pla75]. The construction is closely related to the Bradley–Terry model for paired comparisons [BT52] and to the independent-utility viewpoint of [Thu27]. Probabilistically, it is successive size-biased sampling without replacement; the exponential race representation is discussed in [PT15]. The same distribution is the stationary ranking of the Tsetlin library and has also prompted exact enumerative questions [CDK24].

Cycle counts provide a classical microscopic benchmark. Under the uniform law, the counts of cycles of lengths $1, 2, \dots, L$ converge jointly to independent Poisson variables with means $1, 1/2, \dots, 1/L$; the Ewens family belongs to the same logarithmic-combinatorial framework [ABT03, Tav21]. A nonuniform law need not retain this behavior. In the fixed- q Mallows model, for example, bounded cycle counts can have linear means and Gaussian fluctuations, while the global cycle geometry undergoes a separate localization transition [GP18, H MV23]. Short cycles are therefore a sensitive test of which features of the uniform permutation survive inhomogeneity.

Permutons give the natural global limit theory for permutation sequences [HKM⁺13], but their topology averages away exact lattice coincidences. A complementary theorem in [Muk16] shows that a uniform local approximation to all fixed-dimensional assignment probabilities implies independent Poisson limits for short cycles, with means given by cyclic traces of the limiting density. This theorem explains the form of our constants, but does not subsume our results. Its local hypothesis is stronger than permuton convergence and is not provided by the Luce bulk or consecutive-pattern limits, especially near a singular endpoint. We instead derive microscopic control from the exponential race under the minimal profile hypotheses below, and then determine what replaces the finite-mean Poisson law when an endpoint contributes on every logarithmic scale.

1.2 Finite-mean regime

Because multiplying all weights by the same constant does not change the law, for the finite-mean results we normalize

$$\frac{1}{n} \sum_{i=1}^n \theta_{n,i} = 1. \quad (5)$$

Let f_n be the step profile

$$f_n(x) = \theta_{n,i}, \quad (i-1)/n < x \leq i/n. \quad (6)$$

Assumption 1.1. *There is a measurable $f : (0, 1) \rightarrow (0, \infty)$ such that*

$$\|f_n - f\|_{L^1(0,1)} \rightarrow 0. \quad (7)$$

In particular, $\int_0^1 f = 1$.

Equivalently, under the normalization (5), it is enough to assume $f_n \rightarrow f$ almost everywhere and $\int f = 1$: Scheffe's lemma then gives (7). The integral condition rules out weight mass escaping through a vanishing fraction of labels.

Define, for $t \geq 0$,

$$H(t) := \int_0^1 e^{-tf(u)} du, \quad F(t) := 1 - H(t), \quad D(t) := \int_0^1 f(u) e^{-tf(u)} du. \quad (8)$$

Since $f > 0$ almost everywhere, F is continuous and strictly increasing from 0 to 1. Write $t_x := F^{-1}(x)$ for $0 < x < 1$. The limiting Luce permuton has density

$$\rho(x, y) = \frac{f(x) e^{-f(x)t_y}}{D(t_y)}, \quad (x, y) \in (0, 1)^2. \quad (9)$$

For the finite-mean theorems, the endpoint condition is most naturally stated on logarithmic depth shells. Put

$$k_{n,m} := n - m + 1, \quad 1 \leq m \leq n. \quad (10)$$

For every integer $j \geq 1$, let

$$\mathcal{D}_{n,j} := \left\{ m : j \leq \log \frac{n}{m} < j+1 \right\}. \quad (11)$$

All shell cutoffs J below tend to infinity through integers. When $\mathcal{D}_{n,j}$ is nonempty, define

$$r_{n,j} := \max \mathcal{D}_{n,j}, \quad b_{n,j} := \min_{m \in \mathcal{D}_{n,j}} \theta_{n,k_{n,m}}. \quad (12)$$

Assumption 1.2 (Endpoint shell capacity). *Under the normalization (5),*

$$\lim_{J \rightarrow \infty} \limsup_{n \rightarrow \infty} \sum_{\substack{j \geq J \\ \mathcal{D}_{n,j} \neq \emptyset}} \exp\{-b_{n,j}j\} = 0. \quad (13)$$

Since $j \leq \log(n/r_{n,j}) < j+1$, the condition is unchanged if j in (13) is replaced by $\log(n/r_{n,j})$ or by $j+1$. To see this, split the shells according as $b_{n,j} \leq 1$ or $b_{n,j} > 1$; the change costs only a fixed factor in the first case and a summable geometric error in the second.

Condition (13) allows the terminal rates to vanish and permits different lower envelopes on different logarithmic depth scales. A convenient stronger sufficient condition is the existence of $\gamma > 0$, $\varepsilon_0 > 0$, and n_0 such that

$$\theta_{n,k} \geq \gamma \quad \text{whenever } n \geq n_0 \text{ and } (1 - \varepsilon_0)n \leq k \leq n. \quad (14)$$

Another useful sufficient condition, which permits the terminal rates to vanish, is that for some $\delta > 0$ and J_0 , uniformly over $1 \leq m \leq n$ and all sufficiently large n ,

$$\theta_{n,n-m+1} \geq (1 + \delta) \frac{\log \log(en/m)}{\log(en/m)} \quad \text{whenever } \log(en/m) \geq J_0. \quad (15)$$

Indeed, for $m \in \mathcal{D}_{n,j}$ one has $\log(en/m) \in [j+1, j+2)$. Since $x \mapsto (\log x)/x$ is eventually decreasing, for all sufficiently large j ,

$$b_{n,j}j \geq (1 + \delta) \frac{j}{j+2} \log(j+2) \geq (1 + \delta/2) \log(j+2).$$

Thus the shell costs are bounded by $(j+2)^{-1-\delta/2}$, whose tail is summable, and (15) implies Assumption 1.2. The shell condition is strictly weaker and is the hypothesis used in both finite-mean theorems below.

Theorem 1.3. *Under Assumptions 1.1 and 1.2,*

$$\Xi_n \implies \text{PRM}(\rho(x, x) dx) \quad (16)$$

as random finite point measures on $[0, 1]$. In particular,

$$d_{\text{TV}}(\mathcal{L}(X_n), \text{Poi}(\lambda)) \longrightarrow 0, \quad \lambda := \int_0^1 \rho(x, x) dx < \infty. \quad (17)$$

The conclusion is spatial, not only numerical: it records where the fixed points occur. In particular, it implies the predicted Poisson law for their total number.

For $\ell \geq 1$, define

$$\lambda_\ell := \frac{1}{\ell} \int_{[0,1]^\ell} \rho(x_1, x_2) \rho(x_2, x_3) \cdots \rho(x_\ell, x_1) dx_1 \cdots dx_\ell. \quad (18)$$

Theorem 1.4. *Under Assumptions 1.1 and 1.2, for every fixed L ,*

$$(C_{n,1}, \dots, C_{n,L}) \implies (Z_1, \dots, Z_L), \quad (19)$$

where $Z_1, \dots, Z_L$ are independent and $Z_\ell \sim \text{Poi}(\lambda_\ell)$. The convergence in (19) also holds in total variation on $\mathbb{Z}_+^L$.

When $f \equiv 1$, one has $\rho \equiv 1$ and $\lambda_\ell = 1/\ell$, recovering the classical uniform-permutation law. The factor $1/\ell$ in (18) accounts for the ℓ possible choices of a root on a directed cycle. Both theorems allow a merely measurable limiting profile and highly irregular triangular arrays; no interior continuity or uniform upper and lower bounds are imposed.

1.3 Power-law endpoint regime

The endpoint condition in the Poisson theorem cannot simply be omitted. A power zero at the right endpoint produces cycles near the upper-right corner of the permutation matrix; a sufficiently strong power pole at the left endpoint produces them near the lower-left corner. In either case, every logarithmic distance scale from the corner contributes a constant amount. Adding those scales gives order $\log n$ cycles.

The central limit theorems below are the diverging-mean counterparts of the finite-mean Poisson results. If $\mu_n \rightarrow \mu < \infty$, the corresponding limit is $\text{Poi}(\mu)$. If instead $\mu_n \rightarrow \infty$, then the natural comparison is

$$\frac{\text{Poi}(\mu_n) - \mu_n}{\sqrt{\mu_n}} \Rightarrow \mathcal{N}(0, 1).$$

Our endpoint theorems prove the corresponding CLTs for the actual cycle counts, jointly across cycle lengths, endpoint windows, and active corners. The relevant mean is of order $\log n$ in the power-law regime and, for fixed points, $\log \log n$ at the critical pole.

Assumption 1.5. *For some continuous $f : (0, 1) \rightarrow (0, \infty)$,*

$$\theta_{n,i} = f((i - 1/2)/n).$$

At the right endpoint we allow either a power zero

$$f(1 - s) = c_R s^\beta (1 + O(s^\eta)), \quad \beta, c_R, \eta > 0,$$

or a finite positive limit. At the left endpoint we allow either a nonintegrable power pole

$$f(s) = c_L s^{-\alpha} (1 + O(s^\eta)), \quad \alpha > 1, \quad c_L, \eta > 0,$$

or a finite positive limit. At least one of the two power behaviors is present.

Call an endpoint *active* when the corresponding power behavior is present. Thus f is bounded above and below on every interval separated from the active endpoints, and there are no other singularities.

Near an active corner, transition probabilities depend asymptotically on the ratio of two endpoint distances. Taking logarithms turns this ratio into a difference. The resulting increment density is, for $\gamma, q > 0$,

$$p_{\gamma,q}(w) := \gamma q e^{\gamma w} e^{-q e^{\gamma w}}, \quad \mathfrak{b}_{\gamma,\ell}(q) := \frac{1}{\ell} p_{\gamma,q}^{*\ell}(0). \quad (20)$$

Here $p^{*\ell}$ denotes ℓ -fold convolution. Thus $p_{\gamma,q}^{*\ell}(0)/\ell$ is the return density of a closed ℓ -step path, divided by the number of choices of its root. For an active right or left endpoint, respectively, put

$$q_R := \Gamma(1 + 1/\beta)^\beta, \quad b_{R,\ell} := \mathfrak{b}_{\beta,\ell}(q_R), \quad (21)$$

$$q_L := \Gamma(1 - 1/\alpha)^{-\alpha}, \quad b_{L,\ell} := \mathfrak{b}_{\alpha,\ell}(q_L). \quad (22)$$

Set $b_{\sigma,\ell} = 0$ when endpoint σ is inactive, and put $B_\ell := b_{R,\ell} + b_{L,\ell}$.

The coefficient $b_{\sigma,\ell}$ is the mean number of ℓ -cycles per unit of logarithmic distance at corner σ . In particular, $b_{R,1} = \beta q_R e^{-q_R}$ and $b_{L,1} = \alpha q_L e^{-q_L}$.

Theorem 1.6. *Under Assumption 1.5, for every fixed L ,*

$$\left(\frac{C_{n,\ell} - B_\ell \log n}{\sqrt{B_\ell \log n}} \right)_{\ell=1}^L \implies (Z_1, \dots, Z_L), \quad (23)$$

where $Z_1, \dots, Z_L$ are independent standard normal variables. Moreover, $\mathbb{E}C_{n,\ell} \sim B_\ell \log n$ for every fixed ℓ .

The $\log n$ cycles are not concentrated among the first or last few labels. They are spread uniformly in logarithmic distance from an active endpoint. To make this precise, root a cycle $\mathbf{c}$ at its largest label $v(\mathbf{c})$ and put

$$d_L(\mathbf{c}) := v(\mathbf{c}), \quad d_R(\mathbf{c}) := n + 1 - v(\mathbf{c}).$$

Thus $u_\sigma(\mathbf{c}) := \log d_\sigma(\mathbf{c}) / \log n$ records its location relative to corner σ on a scale from 0 to 1. For $I = (a, b] \subset (0, 1)$, let

$$C_{n,\ell}^\sigma(I) := \#\{\ell\text{-cycles } \mathbf{c} : u_\sigma(\mathbf{c}) \in I\}. \quad (24)$$

For fixed points this is simply the logarithmic location of the point.

Theorem 1.7. *Under the assumptions of Theorem 1.6, fix pairwise disjoint intervals $I_1, \dots, I_J$ whose closures lie in $(0, 1)$, and fix L . Jointly over active corners σ , lengths $1 \leq \ell \leq L$, and $1 \leq j \leq J$,*

$$\frac{C_{n,\ell}^\sigma(I_j) - b_{\sigma,\ell}|I_j| \log n}{\sqrt{b_{\sigma,\ell}|I_j| \log n}} \implies \mathcal{N}(0, 1), \quad (25)$$

with independent limits. Moreover, for every such interval I ,

$$\mathbb{E}C_{n,\ell}^\sigma(I) = b_{\sigma,\ell}|I| \log n + o(\log n). \quad (26)$$

In particular, the expected number of cycles with $n^a < d_\sigma \leq n^b$ is $b_{\sigma,\ell}(b - a) \log n + o(\log n)$. Different logarithmic intervals, cycle lengths, and endpoints fluctuate independently. When both endpoints are active, their means add to the constant $B_\ell = b_{L,\ell} + b_{R,\ell}$ in Theorem 1.6.

This location describes the entire cycle, not only its root. More precisely, for every $\delta > 0$, the expected number of cycles counted above that contain a vertex whose logarithmic distance from the same endpoint differs from $\log d_\sigma(\mathbf{c})$ by more than $\delta \log n$ tends to zero.

At the borderline pole $f(s) \asymp s^{-1}$, only fixed points diverge, and they do so on the smaller scale $\log \log n$.

Theorem 1.8. *Suppose $\theta_{n,i} = f((i - 1/2)/n)$ for a continuous positive f satisfying*

$$f(s) = c_L s^{-1} (1 + O(s^\eta)) \quad (s \downarrow 0)$$

for some $c_L, \eta > 0$, and suppose f has a finite positive limit at 1. Then

$$\frac{C_{n,1} - \log \log n}{\sqrt{\log \log n}} \implies \mathcal{N}(0, 1), \quad \mathbb{E}C_{n,1} \sim \log \log n. \quad (27)$$

For every fixed $\ell \geq 2$, one has $\sup_n \mathbb{E}C_{n,\ell} < \infty$.

Theorem 1.8 is the CLT corresponding to a diverging mean, now with $\mu_n = \log \log n$. The means of the longer cycles remain bounded at the critical exponent.

Section 6 proves these endpoint results directly for the sampled profiles stated above. Its population estimates are derived from the endpoint expansions and the bounds supplied by positivity and continuity away from the active endpoints. In particular, the finite-mean normalization and Assumptions 1.1–1.2 are not imposed in the power-law or critical-pole regimes.

The profile singled out in [BCD25, Section 5.2] gives a concrete instance of the endpoint regime.

Corollary 1.9. *Let π_n be a Luce permutation with the standard Sukhatme weights*

$$\theta_{n,i} = n - i + 1, \quad 1 \leq i \leq n.$$

Put $p(w) := e^w e^{-e^w}$ and $b_\ell := p^{*\ell}(0)/\ell$. For every fixed L ,

$$\left(\frac{C_{n,\ell} - b_\ell \log n}{\sqrt{b_\ell \log n}} \right)_{\ell=1}^L \Rightarrow (Z_1, \dots, Z_L), \quad (28)$$

where the coordinates on the right are independent standard normal random variables, and $\mathbb{E}C_{n,\ell} \sim b_\ell \log n$. The spatial conclusion of Theorem 1.7 also holds, with a single active right corner and coefficients b_ℓ . In particular,

$$\frac{C_{n,1} - e^{-1} \log n}{\sqrt{e^{-1} \log n}} \Rightarrow \mathcal{N}(0, 1), \quad b_2 = K_0(2) \approx 0.113894,$$

where K_0 is the modified Bessel function of the second kind.

Proof. Take $f(x) = 1 - x$ and use the interior grid $i/(n+1)$. Its weights are $(n-i+1)/(n+1)$, so scale invariance gives precisely the Sukhatme law. Lemma 6.12 extends the sampled-profile conclusions to this grid. Here $\beta = 1$ and $q_R = 1$, giving the asserted constants and spatial limit. In particular $p(0) = e^{-1}$, and

$$\frac{1}{2} p^{*2}(0) = \frac{1}{2} \int_{\mathbb{R}} e^{-e^w - e^{-w}} dw = K_0(2). \quad \square$$

Thus the standard Sukhatme model is already in the diverging-mean regime for fixed points, with mean asymptotic to $e^{-1} \log n$. Corollary 1.9 proves the corresponding fixed-point CLT and confirms the behavior suggested by the simulations in [BCD25, Section 5.2]. More generally, when a right power zero and a left nonintegrable power pole are both present, cycles occur near both endpoints and their contributions add.

2 Predictable fixed-point probabilities

We expose the permutation in draw order. At step k , only one event can create a fixed point: label k must still be available and must be chosen next. Its conditional probability is explicit even though the fixed-point indicators themselves are dependent.

Let $\mathcal{F}_{n,k} = \sigma(\pi_n(1), \dots, \pi_n(k))$. Immediately before draw k , define the remaining weight

$$W_{n,k} := \sum_{i=1}^n \theta_{n,i} \mathbf{1}\{R_{n,i} \geq k\}. \quad (29)$$

The conditional chance of a fixed point is

$$p_{n,k} := \mathbb{P}(I_{n,k} = 1 \mid \mathcal{F}_{n,k-1}) = \frac{\theta_{n,k} \mathbf{1}\{R_{n,k} \geq k\}}{W_{n,k}}. \quad (30)$$

Both the numerator indicator and the denominator are $\mathcal{F}_{n,k-1}$ -measurable. The random measure

$$A_n := \sum_{k=1}^n p_{n,k} \delta_{k/n} \quad (31)$$

is the predictable compensator of Ξ_n in draw order. The fixed-point problem is therefore reduced to showing that A_n approaches a deterministic measure and that its largest atom vanishes. The following criterion records this reduction.

Lemma 2.1. *Let $I_{n,k} \in \{0, 1\}$ be adapted, let $x_{n,k}$ be deterministic points in a compact space X , and let $p_{n,k} = \mathbb{E}(I_{n,k} \mid \mathcal{F}_{n,k-1})$. Suppose that for a finite deterministic measure ν on X ,*

$$\sum_k p_{n,k} \delta_{x_{n,k}} \implies \nu \quad \text{in probability}, \quad \max_k p_{n,k} \longrightarrow 0 \quad \text{in probability}. \quad (32)$$

Then $\sum_k I_{n,k} \delta_{x_{n,k}}$ converges to a Poisson random measure with intensity ν .

Proof. Fix a nonnegative continuous g , abbreviate $g_{n,k} = g(x_{n,k})$ and $q_{n,k} = 1 - e^{-g_{n,k}}$. We first stop the array before either its total compensator exceeds K or one conditional probability exceeds $\delta < 1$. Formally, define the predictable indicators recursively by

$$H_{n,k} := \mathbf{1} \left\{ p_{n,k} \leq \delta, \sum_{j < k} H_{n,j} p_{n,j} + p_{n,k} \leq K \right\}.$$

Set $\tilde{I}_{n,k} = H_{n,k} I_{n,k}$ and $\tilde{p}_{n,k} = H_{n,k} p_{n,k}$. Then

$$L_{n,m} := \prod_{k \leq m} \frac{e^{-g_{n,k}} \tilde{I}_{n,k}}{1 - \tilde{p}_{n,k} q_{n,k}}. \quad (33)$$

Each factor has conditional mean one, so $(L_{n,m})_m$ is a nonnegative mean-one martingale. This likelihood martingale replaces the product factorization available for independent indicators: here the future probabilities may depend on the preceding outcomes.

For a single factor $Y_{n,k}$ in (33),

$$\mathbb{E}(Y_{n,k}^2 \mid \mathcal{F}_{n,k-1}) = 1 + \frac{\tilde{p}_{n,k}(1 - \tilde{p}_{n,k})q_{n,k}^2}{(1 - \tilde{p}_{n,k}q_{n,k})^2} \leq \exp(C_{g,\delta} \tilde{p}_{n,k}).$$

Iteration therefore gives

$$\sup_m \mathbb{E} L_{n,m}^2 \leq \exp(C_{g,\delta} K), \quad (34)$$

so the stopped likelihoods are uniformly integrable. Meanwhile,

$$\begin{aligned} \prod_k (1 - \tilde{p}_{n,k} q_{n,k}) &= \exp \left\{ - \sum_k \tilde{p}_{n,k} q_{n,k} + O \left(\max_k \tilde{p}_{n,k} \sum_k \tilde{p}_{n,k} \right) \right\} \\ &\longrightarrow \exp \left\{ - \int (1 - e^{-g}) d\nu \right\} \end{aligned}$$

in probability whenever no term is deleted. By (32), choose $K > \nu(X) + 1$ and any fixed $\delta \in (0, 1)$. The probability that a term is deleted tends to zero. Multiplying the last convergence by the uniformly integrable likelihood in (33), and then removing the stopping, gives

$$\mathbb{E} \exp \left\{ - \sum_k g(x_{n,k}) I_{n,k} \right\} \longrightarrow \exp \left\{ - \int (1 - e^{-g}) d\nu \right\},$$

the Laplace functional of $\text{PRM}(\nu)$. □

3 The interior compensator

On every compact subinterval of $[0, 1)$, a law of large numbers for the exponential race controls both the time of the k th arrival and the total rate still present at that time. This identifies the compensator in the bulk. The endpoint, where relative concentration can fail, is postponed to Section 4.

Let

$$\widehat{F}_n(t) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{E_{n,i} \leq t\}, \quad F_n(t) := \mathbb{E} \widehat{F}_n(t) = \frac{1}{n} \sum_{i=1}^n (1 - e^{-\theta_{n,i}t}), \quad (35)$$

$$\widehat{D}_n(t) := \frac{1}{n} \sum_{i=1}^n \theta_{n,i} \mathbf{1}\{E_{n,i} \geq t\}, \quad D_n(t) := \mathbb{E} \widehat{D}_n(t) = \frac{1}{n} \sum_{i=1}^n \theta_{n,i} e^{-\theta_{n,i}t}. \quad (36)$$

Let $\tau_{n,k}$ be the k th order statistic of the exponential variables. Then $W_{n,k} = n\widehat{D}_n(\tau_{n,k})$ and

$$\mathbf{1}\{R_{n,k} \geq k\} = \mathbf{1}\{E_{n,k} \geq \tau_{n,k}\}. \quad (37)$$

Lemma 3.1. *Under Assumption 1.1, for every finite T ,*

$$\sup_{0 \leq t \leq T} |\widehat{F}_n(t) - F(t)| \longrightarrow 0, \quad \sup_{0 \leq t \leq T} |\widehat{D}_n(t) - D(t)| \longrightarrow 0 \quad (38)$$

in probability. Consequently, for every $\alpha < 1$,

$$\max_{1 \leq k \leq \alpha n} |\tau_{n,k} - t_{k/n}| \longrightarrow 0 \quad \text{in probability} \quad (39)$$

and

$$\max_{1 \leq k \leq \alpha n} \left| \frac{W_{n,k}}{n} - D(t_{k/n}) \right| \longrightarrow 0 \quad \text{in probability.} \quad (40)$$

Proof. First, L^1 convergence implies

$$\frac{1}{n} \max_i \theta_{n,i} \longrightarrow 0. \quad (41)$$

Indeed, the mass of any interval of length $1/n$ under f tends uniformly to zero, while $\|f_n - f\|_1 \rightarrow 0$.

The maps $a \mapsto e^{-ta}$ and $a \mapsto ae^{-ta}$ are uniformly Lipschitz in $a \geq 0$ when t ranges over a compact interval; for the second map one may use $|(1 - ta)e^{-ta}| \leq 1$. Hence $F_n \rightarrow F$ and

$D_n \rightarrow D$ uniformly on $[0, T]$. At a fixed time, Hoeffding's inequality gives concentration of $\widehat{F}_n$, while

$$\text{Var}(\widehat{D}_n(t)) \leq \frac{1}{n^2} \sum_i \theta_{n,i}^2 e^{-\theta_{n,i}t} \leq \frac{\max_i \theta_{n,i}}{n}.$$

A finite grid, monotonicity in t , and uniform continuity of F and D upgrade the fixed-time bounds to (38).

Since F is strictly increasing, uniform convergence of the distribution functions implies uniform convergence of their quantiles on $[0, \alpha]$, which gives (39). Substituting the random quantiles in the second part of (38) gives (40). $\square$

Proposition 3.2. *Under Assumption 1.1, for every $\alpha < 1$ and every continuous $g : [0, \alpha] \rightarrow \mathbb{R}$,*

$$\sum_{k \leq \alpha n} g(k/n) p_{n,k} \longrightarrow \int_0^\alpha g(x) \rho(x, x) dx \quad (42)$$

in probability. Moreover,

$$\max_{k \leq \alpha n} p_{n,k} \longrightarrow 0 \quad \text{in probability.} \quad (43)$$

Proof. Put $t_k = t_{k/n}$ and $B_{n,k} = \mathbf{1}\{E_{n,k} \geq t_k\}$. Since $d_\alpha := \min_{0 \leq t \leq t_\alpha} D(t) > 0$, Lemma 3.1 gives

$$\sum_{k \leq \alpha n} g(k/n) p_{n,k} = \frac{1}{n} \sum_{k \leq \alpha n} \frac{g(k/n) \theta_{n,k} \mathbf{1}\{E_{n,k} \geq \tau_{n,k}\}}{D(t_k)} + o_{\mathbb{P}}(1). \quad (44)$$

We next replace the survival indicator by $B_{n,k}$.

Fix $\eta > 0$. The contribution of $k \leq \eta n$ is bounded by a constant times $n^{-1} \sum_{k \leq \eta n} \theta_{n,k}$, whose limit is $\int_0^\eta f$. For $\eta n < k \leq \alpha n$, t_k is bounded away from zero. On the event in (39), the two indicators can differ only when $E_{n,k}$ lies in a shrinking interval around t_k . Uniformly on this range,

$$\theta_{n,k} \mathbb{P}(|E_{n,k} - t_k| \leq \delta) \leq \int_{t_k - \delta}^{t_k + \delta} \theta_{n,k}^2 e^{-\theta_{n,k}s} ds \leq C_\eta \delta.$$

First send $n \rightarrow \infty$, then $\delta \downarrow 0$, and finally $\eta \downarrow 0$. This proves that the replacement costs $o_{\mathbb{P}}(1)$.

The variables $B_{n,k}$ are independent. On $k > \eta n$, the variance of the resulting normalized sum is $O_\eta(n^{-1})$, because $\sup_{a \geq 0} a^2 e^{-at_\eta} < \infty$. The initial ηn terms are again controlled by $n^{-1} \sum_{k \leq \eta n} \theta_{n,k}$. Therefore the sum is asymptotic in probability to its expectation. It remains to identify that expectation. On $[\eta, \alpha]$, view the sum as the integral of a step function and put $k_n(x) = \lceil nx \rceil$. Uniformly there, $t_{k_n(x)}/n \rightarrow t_x$. The map $a \mapsto ae^{-at}$ is globally one-Lipschitz, while, for $t \geq t_\eta/2$, $\sup_{a \geq 0} a^2 e^{-at} < \infty$. Hence the uniform convergence of the times, the continuity of g and D , and $\|f_n - f\|_1 \rightarrow 0$ give

$$\begin{aligned} \frac{1}{n} \sum_{k \leq \alpha n} \frac{g(k/n) \theta_{n,k} e^{-\theta_{n,k} t_k}}{D(t_k)} &\longrightarrow \int_0^\alpha \frac{g(x) f(x) e^{-f(x) t_x}}{D(t_x)} dx \\ &= \int_0^\alpha g(x) \rho(x, x) dx. \end{aligned}$$

The discarded initial block is bounded by $C_\alpha \int_0^\eta f_n(x) dx$; it vanishes after first taking $n \rightarrow \infty$ and then $\eta \downarrow 0$. This also handles $x = 0$.

For the maximum, on the high-probability event from Lemma 3.1,

$$\max_{k \leq \alpha n} p_{n,k} \leq \frac{2 \max_i \theta_{n,i}}{nd_\alpha} = o(1)$$

by (41). □

Corollary 3.3. *For every $\alpha < 1$,*

$$\Xi_n|_{[0,\alpha]} \implies \text{PRM}(\rho(x, x) \mathbf{1}_{[0,\alpha]}(x) dx). \quad (45)$$

Proof. Apply Lemma 2.1 and Proposition 3.2. □

4 The last few positions

It remains to rule out fixed points in the last $o(n)$ positions, uniformly enough to let $\alpha \uparrow 1$. Directly concentrating the remaining total weight is fragile: near the end its expectation may be tiny, so an absolute concentration estimate gives no useful relative error. We instead condition on the ringing time of the candidate label and count how many other clocks survive. The argument has no random denominator.

For this section only, suppress the first subscript n . Let

$$S(t) := \sum_{i=1}^n e^{-\theta_i t} = \mathbb{E} \sum_{i=1}^n \mathbf{1}\{E_i > t\}. \quad (46)$$

Lemma 4.1. *For every label k ,*

$$\mathbb{P}(R_k = k) = \int_0^\infty \theta_k e^{-\theta_k t} \mathbb{P} \left(\sum_{j \neq k} \mathbf{1}\{E_j > t\} = n - k \right) dt. \quad (47)$$

Proof. Condition on $E_k = t$. The rank is k exactly when $n - k$ of the other exponential variables are larger than t . □

For $m = 1, \dots, n$, write $k_m = n - m + 1$ and put

$$N(t) := \sum_{j=1}^n \mathbf{1}\{E_j > t\}, \quad N_{-k}(t) := \sum_{j \neq k} \mathbf{1}\{E_j > t\}.$$

For $x > 0$, also write

$$\mathcal{Q}(x) := -\log(1 - e^{-x}). \quad (48)$$

The following block estimate is the basic endpoint-capacity inequality. It uses the actual survivor curve S , and therefore can be substantially stronger than a bound obtained only from the mean-one normalization.

Proposition 4.2 (Block endpoint capacity). *Let $\mathcal{B} \subseteq [n]$ be a nonempty set of terminal depths, and put*

$$r_{\mathcal{B}} := \max \mathcal{B}, \quad a_{\mathcal{B}} := \min_{m \in \mathcal{B}} \theta_{k_m}.$$

For $t_{\mathcal{B}} > 0$, set

$$B_{\mathcal{B}} := S(t_{\mathcal{B}}) - 1.$$

If $B_{\mathcal{B}} > r_{\mathcal{B}}$, then

$$\mathbb{E} \sum_{m \in \mathcal{B}} I_{k_m} \leq |\mathcal{B}| \exp \left\{ -\frac{(B_{\mathcal{B}} - r_{\mathcal{B}})^2}{2B_{\mathcal{B}}} \right\} + \mathcal{Q}(a_{\mathcal{B}} t_{\mathcal{B}}). \quad (49)$$

Proof. For $0 \leq t \leq t_{\mathcal{B}}$ and $m \in \mathcal{B}$,

$$\mathbb{E} N_{-k_m}(t) \geq S(t) - 1 \geq B_{\mathcal{B}}.$$

Since $m - 1 < r_{\mathcal{B}} < B_{\mathcal{B}}$, the Chernoff lower-tail bound gives

$$\mathbb{P}(N_{-k_m}(t) = m - 1) \leq \exp \left\{ -\frac{(B_{\mathcal{B}} - r_{\mathcal{B}})^2}{2B_{\mathcal{B}}} \right\}.$$

Lemma 4.1, followed by integration of the exponential density, bounds the total contribution from $t \leq t_{\mathcal{B}}$ by the first term in (49).

For the remaining times, independence gives

$$\begin{aligned} \mathbb{P}(N(t) = m - 1) &\geq \mathbb{P}(E_{k_m} \leq t, N_{-k_m}(t) = m - 1) \\ &= (1 - e^{-\theta_{k_m} t}) \mathbb{P}(N_{-k_m}(t) = m - 1). \end{aligned}$$

Consequently,

$$\theta_{k_m} e^{-\theta_{k_m} t} \mathbb{P}(N_{-k_m}(t) = m - 1) \leq \frac{\theta_{k_m}}{e^{\theta_{k_m} t} - 1} \mathbb{P}(N(t) = m - 1).$$

For fixed $t > 0$, the map $a \mapsto a/(e^{at} - 1)$ is decreasing. Moreover, the integers $m - 1$, $m \in \mathcal{B}$, are distinct. Hence the sum of the last display over $m \in \mathcal{B}$ is at most

$$\frac{a_{\mathcal{B}}}{e^{a_{\mathcal{B}} t} - 1} \sum_{m \in \mathcal{B}} \mathbb{P}(N(t) = m - 1) \leq \frac{a_{\mathcal{B}}}{e^{a_{\mathcal{B}} t} - 1}.$$

Integrating from $t_{\mathcal{B}}$ to infinity gives

$$\int_{t_{\mathcal{B}}}^{\infty} \frac{a_{\mathcal{B}}}{e^{a_{\mathcal{B}} t} - 1} dt = -\log(1 - e^{-a_{\mathcal{B}} t_{\mathcal{B}}}),$$

which proves the claim. $\square$

There is a complementary individual estimate for a label whose very small rate is badly aligned with its prescribed terminal depth. Let $A_{m-1}^{(-k_m)}$ denote the sum of the $m - 1$ smallest rates among $\{\theta_i : i \neq k_m\}$, with $A_0^{(-k_1)} = 0$.

Lemma 4.3 (Individual slow-label charge). *For every m ,*

$$\mathbb{P}(R_{k_m} = k_m) \leq \frac{\theta_{k_m}}{\theta_{k_m} + A_{m-1}^{(-k_m)}}. \quad (50)$$

Proof. Immediately before the draw at which m clocks remain, condition on the surviving set. If it contains k_m , the conditional chance that k_m rings next is its rate divided by the total surviving rate. The other $m - 1$ survivors have total rate at least $A_{m-1}^{(-k_m)}$. Since a fixed point at k_m requires both survival to this stage and selection at the next draw, taking expectations proves (50). $\square$

For a common block $m = 1, \dots, M$, the key deterministic observation is

$$\sum_{m=1}^M \mathbf{1}\{N_{-k_m}(t) = m - 1\} \leq 2. \quad (51)$$

Indeed, if $N(t) = \sum_j \mathbf{1}\{E_j > t\} = s$, then the displayed event can hold only for $m = s$ when $E_{k_m} > t$, or for $m = s + 1$ when $E_{k_m} \leq t$. Thus, at any fixed time, at most two terminal labels can have the correct survivor count.

Proposition 4.4. *Assume (5). Let $1 \leq M \leq \varepsilon_0 n$, and suppose $\theta_{k_m} \geq \gamma$ for $1 \leq m \leq M$. Put*

$$B := \sqrt{nM}, \quad (52)$$

and let s be determined by $S(s) = B$. If $M \leq (B - 1)/2$ and $s \geq 1/\gamma$, then for a universal $c > 0$,

$$\mathbb{E} \sum_{m=1}^M I_{k_m} \leq M e^{-cB} + 2e^{-\gamma s} \leq M e^{-cB} + 2 \left(\frac{2B}{n} \right)^{\gamma/2}. \quad (53)$$

Consequently,

$$\limsup_{n \rightarrow \infty} \mathbb{E} \sum_{k > (1-\varepsilon)n} I_{n,k} \leq 2^{1+\gamma/2} \varepsilon^{\gamma/4} \quad (54)$$

for all sufficiently small $\varepsilon > 0$.

Proof. For $t \leq s$ and every $m \leq M$,

$$\mathbb{E} N_{-k_m}(t) \geq S(t) - 1 \geq B - 1 \geq 2M.$$

A Chernoff lower-tail bound for a sum of independent Bernoulli variables gives

$$\mathbb{P}(N_{-k_m}(t) = m - 1) \leq e^{-cB}.$$

Using Lemma 4.1 and integrating the exponential density, the total contribution of $t \leq s$ is at most $M e^{-cB}$.

For $t \geq s \geq 1/\gamma$, the map $a \mapsto a e^{-at}$ is decreasing on $[\gamma, \infty)$. Hence, by (51),

$$\begin{aligned} \sum_{m=1}^M \int_s^\infty \theta_{k_m} e^{-\theta_{k_m} t} \mathbb{P}(N_{-k_m}(t) = m - 1) dt \\ \leq \int_s^\infty \gamma e^{-\gamma t} \sum_{m=1}^M \mathbb{P}(N_{-k_m}(t) = m - 1) dt \leq 2e^{-\gamma s}. \end{aligned}$$

It remains to bound s . The normalization implies that at least $n/2$ weights are at most 2. Thus

$$S(t) \geq \frac{n}{2} e^{-2t}.$$

Since $S(s) = B$, we get $s \geq \frac{1}{2} \log(n/(2B))$, proving the second inequality in (53).

Finally take $M = \lceil \varepsilon n \rceil$. Then $B/n \rightarrow \sqrt{\varepsilon}$, the first term in (53) vanishes, and the required side conditions hold when ε is sufficiently small. This gives (54). $\square$

The particular survivor buffer used in the preceding argument is convenient but not intrinsic. A square-root buffer will be useful for both fixed points and short cycles.

Lemma 4.5 (Diagonal shell buffer). *Under Assumption 1.2, the choice $h_j = \sqrt{j}$ satisfies*

$$\lim_{J \rightarrow \infty} \limsup_{n \rightarrow \infty} \sum_{\substack{j \geq J \\ \mathcal{D}_{n,j} \neq \emptyset}} \mathcal{Q}(b_{n,j}(j - h_j)) = 0. \quad (55)$$

Moreover, $h_j \uparrow \infty$, $h_j < j/4$ eventually, and $j - h_j$ is nondecreasing.

Proof. Write $x_{n,j} = b_{n,j}j$ and omit empty shells. Put

$$y_{n,j} := b_{n,j}(j - h_j) = x_{n,j} \left(1 - \frac{1}{\sqrt{j}}\right).$$

If $x_{n,j} \leq \sqrt{j}$, then $y_{n,j} \geq x_{n,j} - 1$ and hence $e^{-y_{n,j}} \leq ee^{-x_{n,j}}$. Otherwise, $y_{n,j} > \sqrt{j} - 1$. Assumption 1.2 therefore gives

$$\lim_{J \rightarrow \infty} \limsup_n \sum_{j \geq J} e^{-y_{n,j}} \leq e \lim_{J \rightarrow \infty} \limsup_n \sum_{j \geq J} e^{-x_{n,j}} + \lim_{J \rightarrow \infty} \sum_{j \geq J} e^{1-\sqrt{j}} = 0.$$

The same estimate shows that $y_{n,j}$ tends uniformly to infinity in the iterated tail. Hence $\mathcal{Q}(y_{n,j}) \leq 2e^{-y_{n,j}}$ there, proving (55). The remaining assertions are immediate from $h_j = \sqrt{j}$. $\square$

Proposition 4.6 (Shell endpoint tightness). *Under the normalization (5) and Assumption 1.2,*

$$\lim_{J \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{E} \sum_{m \leq ne^{-J}} I_{n,k_{n,m}} = 0. \quad (56)$$

Equivalently,

$$\lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} \mathbb{E} \sum_{k > (1-\varepsilon)n} I_{n,k} = 0. \quad (57)$$

Proof. Let h_j be supplied by Lemma 4.5. For a nonempty shell $\mathcal{D}_{n,j}$, apply Proposition 4.2 with

$$\mathcal{B} = \mathcal{D}_{n,j}, \quad t_{\mathcal{B}} = \log \frac{n}{r_{n,j}} - h_j.$$

For all sufficiently large j this time is positive. Jensen's inequality and (5) give

$$S(t_{\mathcal{B}}) \geq ne^{-t_{\mathcal{B}}} = r_{n,j}e^{h_j}.$$

It follows that $B_{\mathcal{B}} \geq r_{n,j}e^{h_j} - 1 > r_{n,j}$ for all sufficiently large j , so the hypothesis of Proposition 4.2 holds. Moreover, for a universal $c > 0$,

$$\frac{(B_{\mathcal{B}} - r_{n,j})^2}{2B_{\mathcal{B}}} \geq cr_{n,j}e^{h_j}.$$

Thus the first term in (49) is at most

$$|\mathcal{D}_{n,j}|e^{-cr_{n,j}e^{h_j}} \leq r_{n,j}e^{-cr_{n,j}e^{h_j}}.$$

The nonempty shells have distinct maxima. Since h_j is nondecreasing,

$$\sum_{j \geq J} r_{n,j} e^{-cr_{n,j} e^{h_j}} \leq \sum_{r \geq 1} r e^{-cre^{h_J}} \longrightarrow 0 \quad (J \rightarrow \infty),$$

uniformly in n . The sum of the second terms in (49) tends to zero by (55), because $t_B \geq j - h_j$ and $\mathcal{Q}$ is decreasing. Since the union of the shells $j \geq J$ is precisely the set of depths with $m \leq ne^{-J}$, this proves (56); the equivalent formulation (57) follows by monotonicity. $\square$

Corollary 4.7 (Exceptional-shell refinement). *Assume the normalization (5). For each n , let $\mathcal{E}_n \subseteq [n]$ be a set of exceptional terminal depths and put*

$$\mathcal{G}_{n,j} := \mathcal{D}_{n,j} \setminus \mathcal{E}_n.$$

When $\mathcal{G}_{n,j}$ is nonempty, define

$$r_{n,j}^G := \max \mathcal{G}_{n,j}, \quad b_{n,j}^G := \min_{m \in \mathcal{G}_{n,j}} \theta_{n,k_{n,m}}.$$

Let $A_{n,m-1}^{(-k_{n,m})}$ be the sum of the $m-1$ smallest rates after deleting $\theta_{n,k_{n,m}}$. Suppose

$$\lim_{J \rightarrow \infty} \limsup_{n \rightarrow \infty} \left[\sum_{\substack{j \geq J \\ \mathcal{G}_{n,j} \neq \emptyset}} e^{-b_{n,j}^G j} + \sum_{\substack{m \in \mathcal{E}_n \\ m \leq ne^{-J}}} \frac{\theta_{n,k_{n,m}}}{\theta_{n,k_{n,m}} + A_{n,m-1}^{(-k_{n,m})}} \right] = 0. \quad (58)$$

Then the endpoint conclusions (56)–(57) hold. Consequently, Theorem 1.3 remains valid if Assumption 1.2 is replaced by (58).

Proof. Apply the proof of Lemma 4.5 to the regular blocks $\mathcal{G}_{n,j}$. In the proof of Proposition 4.6, replace $\mathcal{D}_{n,j}$, $r_{n,j}$, and $b_{n,j}$ by their regular-block versions and use the cutoff $\log(n/r_{n,j}^G) - h_j$. This cutoff is at least $j - h_j$, so the same decreasing- $\mathcal{Q}$ bound makes the regular contribution vanish. Lemma 4.3 bounds the exceptional contribution by the second sum in (58). Adding the two proves the endpoint bounds. These bounds replace each invocation of Proposition 4.6 in the proof of Theorem 1.3, which is otherwise unchanged. $\square$

Remark 4.8. The uniform terminal support condition (14) implies Assumption 1.2. Indeed, for all sufficiently large j , every label in $\mathcal{D}_{n,j}$ lies in that terminal interval, so $b_{n,j} \geq \gamma$ and the shell sum is bounded by the geometric tail $\sum_{j \geq J} e^{-\gamma j}$. Proposition 4.4 records the corresponding direct estimate.

At the shell level, (15) can be weakened further. For example, it is enough that, uniformly on all sufficiently late nonempty shells,

$$b_{n,j} j \geq \log j + (1 + \delta) \log \log j \quad (59)$$

for some $\delta > 0$, because the shell costs are then bounded by $1/(j(\log j)^{1+\delta})$. In terms of an individual terminal depth with $q = \log(n/m)$, the corresponding envelope has order

$$\theta_{n,n-m+1} \geq \frac{\log q + (1 + \delta) \log \log q}{q}.$$

The usual iterated-log refinements of the convergent Bertrand series give still weaker sufficient envelopes, and sparse bad shells need only have summable costs. Provided the remaining shell

costs already have uniformly vanishing tails, a single moving anomalous shell is allowed whenever its product $b_{n,j}j$ tends to infinity.

The minimum $b_{n,j}$ can nevertheless be spoiled by one harmless ultra-slow label. Corollary 4.7 removes such labels from their regular shells and charges them individually. This exception mechanism is proved here only for fixed points. The short-cycle argument below uses the lower floor on every label in a shell, so no analogue under the exceptional-shell refinement is claimed.

Proof of Theorem 1.3. For fixed $\alpha < 1$, Corollary 3.3 gives the desired Poisson limit on $[0, \alpha]$. Proposition 4.6 and Markov's inequality imply

$$\lim_{\alpha \uparrow 1} \limsup_{n \rightarrow \infty} \mathbb{P}(\Xi_n((\alpha, 1]) > 0) = 0. \quad (60)$$

It remains to verify that the proposed limiting intensity is finite; weak convergence alone does not imply convergence of means. Fix $\alpha < \beta < 1$. Interior count convergence, followed by truncation and Fatou's lemma, gives

$$\begin{aligned} \int_{\alpha}^{\beta} \rho(x, x) dx &= \mathbb{E} \text{Poi} \left(\int_{\alpha}^{\beta} \rho(x, x) dx \right) \\ &\leq \liminf_{n \rightarrow \infty} \mathbb{E} \Xi_n((\alpha, \beta]) \leq \limsup_{n \rightarrow \infty} \mathbb{E} \Xi_n((\alpha, 1]). \end{aligned}$$

Letting $\beta \uparrow 1$ and using monotone convergence gives

$$\int_{\alpha}^1 \rho(x, x) dx \leq \limsup_{n \rightarrow \infty} \mathbb{E} \Xi_n((\alpha, 1]).$$

Proposition 4.6 shows that the right side tends to zero as $\alpha \uparrow 1$, first along $\alpha = 1 - e^{-J}$ and then for arbitrary α by monotonicity. Together with finiteness of the intensity on every compact subinterval of $[0, 1)$, this proves both

$$\lambda := \int_0^1 \rho(x, x) dx < \infty, \quad \int_{\alpha}^1 \rho(x, x) dx \longrightarrow 0. \quad (61)$$

Now let $\alpha \uparrow 1$ in the interior point-process limit. Equations (60) and (61) give the full convergence (16), and hence $X_n \Rightarrow \text{Poi}(\lambda)$.

For integer-valued random variables, convergence in distribution to a proper integer-valued law implies pointwise convergence of the probability mass functions; Scheffe's lemma then upgrades this to total variation convergence, giving (17). $\square$

5 Short cycles

This section proves Theorem 1.4. There are again two tasks. In the bulk, a linear reservoir of moderate-rate clocks yields the finite-dimensional local law needed for factorial moments. Near the endpoint, the fixed-point estimate no longer suffices: the label that closes a late cycle may come from anywhere. We handle this by inserting every proposed finite path into a common background permutation. The construction retains the essential permutation constraint that each rank has one preimage.

For an integer m , write $[m]_{\neq}^r$ for the set of ordered r -tuples of distinct elements of $[m]$.

Lemma 5.1. *Under Assumption 1.1, for every $\alpha < 1$ there are $d, \eta > 0$ such that, for all sufficiently large n ,*

$$\#\{i : \theta_{n,i} \geq d\} \geq (\alpha + 2\eta)n. \quad (62)$$

Consequently, after deleting a fixed number of clocks and before αn arrivals, the remaining total rate is at least $d\eta n$.

Proof. Since $f > 0$ almost everywhere, choose $d, \eta > 0$ so that $|\{x : f(x) \geq 2d\}| > \alpha + 3\eta$. The part of this set on which $f_n < d$ has measure at most $d^{-1}\|f_n - f\|_1$, which proves (62). At most $\alpha n + O(1)$ members of the reservoir have arrived or been deleted, leaving at least ηn clocks of rate at least d . $\square$

Lemma 5.2. *Under Assumption 1.1, fix $r \geq 1$, $\alpha < 1$, a permutation $\tau \in S_r$, and a continuous function $g : [0, \alpha]^r \rightarrow \mathbb{R}$. Then*

$$\begin{aligned} \sum_{\mathbf{i} \in [[\alpha n]]^r_{\neq}} g(\mathbf{i}/n) \mathbb{P}(R_{n,i_a} = i_{\tau(a)}, 1 \leq a \leq r) \\ \longrightarrow \int_{[0,\alpha]^r} g(\mathbf{x}) \prod_{a=1}^r \rho(x_a, x_{\tau(a)}) d\mathbf{x}. \end{aligned} \quad (63)$$

Moreover, uniformly over distinct marked labels $i_1, \dots, i_r$ and distinct ranks $j_1, \dots, j_r \leq \alpha n$,

$$\mathbb{P}(R_{n,i_a} = j_a, 1 \leq a \leq r) \leq K_{r,\alpha} n^{-r} \prod_{a=1}^r \theta_{n,i_a}. \quad (64)$$

Proof. Delete the marked clocks and denote the unmarked order statistics by $T_0 = 0 < T_1 < \dots < T_{n-r}$. If W_q° is the remaining unmarked rate, the memoryless property gives

$$T_{q+1} - T_q = \frac{\xi_q}{W_q^\circ}, \quad (65)$$

where, conditionally on the elimination chain, the ξ_q are independent standard exponentials. Lemma 5.1 gives $W_q^\circ \geq d\eta n$ for $q \leq \alpha n$. Sorting the prescribed ranks, and grouping marked clocks which must enter the same unmarked gap, therefore gives

$$\mathbb{P}(R_{n,i_a} = j_a, a \leq r) \leq K_{r,\alpha} n^{-r} \prod_{a=1}^r \theta_{n,i_a},$$

which is (64).

We identify the asymptotic first with all marked rates bounded by a fixed M . Deleting finitely many such clocks changes the empirical race by $o(1)$, uniformly in their labels. Lemma 3.1 and (65) then show, uniformly when the prescribed ranks are separated by at least ζn , that

$$n^r \mathbb{P}(R_{n,i_a} = j_a, a \leq r) - \prod_{a=1}^r \frac{\theta_{n,i_a} e^{-\theta_{n,i_a} t_{j_a/n}}}{D(t_{j_a/n})} \longrightarrow 0. \quad (66)$$

Indeed, each marked clock is inserted into a relevant unmarked gap; Taylor expansion of its exponential density is uniform for rates at most M , and the rescaled separated gaps converge jointly to independent standard exponentials.

The contribution to (63) of tuples containing a rate larger than M is, by (64), at most a constant times

$$\sum_{a=1}^r \left(\frac{1}{n} \sum_i \theta_{n,i} \mathbf{1}_{\{\theta_{n,i} > M\}} \right) \left(\frac{1}{n} \sum_i \theta_{n,i} \right)^{r-1}. \quad (67)$$

It vanishes as first $n \rightarrow \infty$ and then $M \rightarrow \infty$. After this rate truncation, tuples with two coordinates within ζn have contribution $O_{r,M}(\zeta)$; the discarded high-rate part is still controlled by (67).

It remains to pass from the grid kernel in (66) to ρ . On rates at most M , the kernel is bounded and uniformly continuous in its rank coordinate on $[0, \alpha]$, so the grid sum differs from the corresponding step integral by $o_M(1)$. The map $a \mapsto ae^{-at}$ is one-Lipschitz, uniformly for $a, t \geq 0$, and $D(t_y)$ is bounded below for $y \leq \alpha$. A telescoping product and $\|f_n - f\|_1 \rightarrow 0$ identify the integral. The parts on which either f_n or f exceeds M are bounded by their L^1 tails, because every numerator is at most its source rate. Let $n \rightarrow \infty$, then $\zeta \downarrow 0$, and finally $M \rightarrow \infty$ to obtain (63). $\square$

We now set up the endpoint resampling argument. Fix a proposed ℓ -cycle and abbreviate $\theta_{n,i}$ to θ_i . Generate a background family $E_i^0 \sim \text{Exp}(\theta_i)$, with order statistics

$$0 = T_0 < T_1 < \dots < T_n < T_{n+1} = \infty,$$

and replace the clocks on the proposed cycle by independent copies. Put

$$J_j := (T_{(j-\ell) \vee 0}, T_{(j+\ell) \wedge (n+1)}), \quad p_{i,j} := \int_{J_j} \theta_i e^{-\theta_i t} dt. \quad (68)$$

The window J_j contains every time that could acquire rank j after these ℓ replacements. Since replacing ℓ clocks changes every rank by at most ℓ , averaging over the background gives

$$\mathbb{P}(R_{i_a} = j_a, 1 \leq a \leq \ell) \leq \mathbb{E} \prod_{a=1}^{\ell} p_{i_a, j_a}. \quad (69)$$

Each time belongs to at most $2\ell + 1$ windows, and consequently

$$\sum_{j=1}^n p_{i,j} \leq 2\ell + 1. \quad (70)$$

The next lemma iterates this insertion bound along a finite path. Its distinct-source condition is exactly the condition supplied by a genuine cycle.

Lemma 5.3. *For fixed $\ell, m \geq 1$ and every $v \in [n]$,*

$$\mathbb{E} \sum_{\substack{u_0, \dots, u_{m-1} \in [n] \\ u_0, \dots, u_{m-1} \text{ distinct}}} p_{u_0, u_1} \cdots p_{u_{m-2}, u_{m-1}} p_{u_{m-1}, v} \leq K_{\ell, m}. \quad (71)$$

Proof. Generate one independent replacement E_i^1 for every background clock. For a fixed source sequence $\mathbf{u}$, the product in (71) is the conditional probability that every $E_{u_a}^1$ falls in the window indexed by the next vertex, ending with J_v . Swap $E_{u_a}^0$ with $E_{u_a}^1$ for all a . These swaps concern

distinct labels and preserve the joint law. If $\tilde{R}$ is the rank permutation of the new background, success before the swap implies

$$|\tilde{R}_{u_a} - u_{a+1}| \leq q_{\ell,m} \quad (a < m-1), \quad |\tilde{R}_{u_{m-1}} - v| \leq q_{\ell,m}, \quad (72)$$

where $q_{\ell,m} = \ell + m + 2$ is sufficient: membership in an old window costs at most $\ell + 1$ ranks, and the m swaps cost at most another m .

For a deterministic permutation σ , give u an edge to every w with $|\sigma(u) - w| \leq q_{\ell,m}$. Each target has indegree at most $2q_{\ell,m} + 1$, since every rank has one preimage. Thus at most $(2q_{\ell,m} + 1)^m$ length- m paths end at v . Sum the measure-preserving comparison over $\mathbf{u}$. $\square$

Put $c_j = \sum_k p_{k,j}$. Expanding this sum and separating whether its source is new gives the useful consequence

$$\sup_{v \in [n]} \mathbb{E} \sum_{\substack{i_2, \dots, i_\ell \in [n] \setminus \{v\} \\ i_2, \dots, i_\ell \text{ distinct}}} c_{i_2} p_{i_2, i_3} \cdots p_{i_\ell, v} \leq K_\ell. \quad (73)$$

Indeed, when the source k in $c_{i_2} = \sum_k p_{k, i_2}$ is distinct from $i_2, \dots, i_\ell$, Lemma 5.3 applies. If $k = i_a$, discard the extra factor $p_{i_a, i_2} \leq 1$ and apply the lemma to the remaining path. The interpretation for $\ell = 1$ is $\sup_v \mathbb{E} c_v < \infty$.

We first prepare two truncations: high rates globally and low rates on compact label intervals. For $M > 1$ and $\delta > 0$, let

$$H_{n,M} := \{i : \theta_{n,i} > M\}, \quad L_{n,\delta} := \{i : \theta_{n,i} < \delta\},$$

and let $V_{n,L}(S)$ be the number of vertices in S which belong to cycles of length at most L .

Proposition 5.4 (Rate truncation). *Under Assumption 1.1,*

$$\lim_{M \rightarrow \infty} \limsup_n \mathbb{E} V_{n,L}(H_{n,M}) = 0, \quad (74)$$

$$\lim_{\delta \downarrow 0} \limsup_n \mathbb{E} V_{n,L}(L_{n,\delta} \cap [\lfloor \beta n \rfloor]) = 0 \quad (\beta < 1). \quad (75)$$

Proof. The L^1 profile convergence gives

$$\lim_{M \rightarrow \infty} \limsup_n \frac{1}{n} \sum_{i \in H_{n,M}} \theta_{n,i} = 0, \quad \lim_{\delta \downarrow 0} \limsup_n \frac{|L_{n,\delta}|}{n} = 0. \quad (76)$$

For the second relation, compare $\{f_n < \delta\}$ with $\{f < 2\delta\}$ and use $f > 0$ almost everywhere.

Choose M so large that the labels of rate at most M carry total weight at least $n/2$. If $\theta_i > M$, then, with $G_n(t) = \sum_k \theta_k e^{-\theta_k t}$,

$$\theta_i e^{-\theta_i t} \leq \frac{2\theta_i}{n} G_n(t).$$

Indeed, all rates at most M have survival factor at least $e^{-\theta_i t}$. Since $c_j = \int_{J_j} G_n(t) dt$, this gives

$$p_{i,j} \leq \frac{2\theta_i}{n} c_j. \quad (77)$$

Start a proposed cycle at $i \in H_{n,M}$, apply (69) and then (77) to its first edge. For $\ell \geq 2$,

$$\begin{aligned} \mathbb{P}(i \text{ belongs to an } \ell\text{-cycle}) &\leq \mathbb{E} \sum_{\substack{i_2, \dots, i_\ell \neq i \\ i_2, \dots, i_\ell \text{ distinct}}} p_{i,i_2} p_{i_2,i_3} \cdots p_{i_\ell,i} \\ &\leq \frac{2\theta_i}{n} \mathbb{E} \sum_{\substack{i_2, \dots, i_\ell \neq i \\ i_2, \dots, i_\ell \text{ distinct}}} c_{i_2} p_{i_2,i_3} \cdots p_{i_\ell,i} \leq K_\ell \frac{\theta_i}{n} \end{aligned}$$

by (73). For fixed points, use $p_{i,i} \leq 2\theta_i c_i/n$ and the $\ell = 1$ interpretation of that estimate. Summing over i and $\ell \leq L$ gives

$$\mathbb{E} V_{n,L}(H_{n,M}) \leq \frac{K_L}{n} \sum_{i \in H_{n,M}} \theta_{n,i}. \quad (78)$$

This proves (74). Now fix $\beta < 1$. Lemma 5.1, applied at a fixed number larger than β , and (65) give

$$\sup_{i \leq \beta n} \mathbb{E}|J_i| \leq K_L/n. \quad (79)$$

In a cycle avoiding $H_{n,M}$, the edge closing into i is at most $M|J_i|$. Sum all other edges with (70) to get

$$\mathbb{P}(i \text{ lies in a short cycle avoiding } H_{n,M}) \leq K_{L,M}/n.$$

A cycle meeting both sets contains at most L vertices from $L_{n,\delta} \cap [\lfloor \beta n \rfloor]$ per high-rate vertex. Therefore

$$\mathbb{E} V_{n,L}(L_{n,\delta} \cap [\lfloor \beta n \rfloor]) \leq L \mathbb{E} V_{n,L}(H_{n,M}) + K_{L,M} \frac{|L_{n,\delta}|}{n}.$$

Use (76), then let $M \rightarrow \infty$, to prove (75). $\square$

For $\alpha < 1$, let $C_{n,\ell}^{(\alpha)}$ count cycles all of whose labels are at most αn .

Proposition 5.5 (Short-cycle shell tightness). *Under Assumptions 1.1 and 1.2, for every fixed L ,*

$$\lim_{\alpha \uparrow 1} \limsup_n \mathbb{E} \sum_{\ell=1}^L (C_{n,\ell} - C_{n,\ell}^{(\alpha)}) = 0. \quad (80)$$

Proof. The case $\ell = 1$ is Proposition 4.6. Fix $2 \leq \ell \leq L$ and put $B = 2\ell + 1$. We first restrict to cycles avoiding $H_{n,M}$. Proposition 5.4 shows that the discarded contribution vanishes as $M \rightarrow \infty$.

Fix a large shell cutoff J_0 and then $\beta < 1$ so close to one that, for all sufficiently large n , every label larger than βn belongs to a shell with index at least J_0 . We also discard cycles meeting $L_{n,\delta} \cap [\lfloor \beta n \rfloor]$. Their contribution vanishes as $\delta \downarrow 0$ by Proposition 5.4. Thus every remaining cycle vertex has rate at most M , and every remaining vertex at most βn has rate at least δ .

Rotate a proposed cycle uniquely as

$$u \longrightarrow v \longrightarrow i_3 \longrightarrow \cdots \longrightarrow i_\ell \longrightarrow u, \quad v = \max\{u, v, i_3, \dots, i_\ell\}.$$

Conditional on the common background in (68), let

$$A_{v,u} := \sum_{i_3, \dots, i_\ell} p_{v,i_3} \cdots p_{i_\ell,u},$$

where the intermediate sums range over retained labels; both distinctness and the genuine-cycle restrictions $i_a < v$ are dropped, while the outer pair remains restricted by $u < v$. For $\ell = 2$ this means $A_{v,u} = p_{v,u}$. Iterating (70) gives the pointwise bound

$$\sum_u A_{v,u} \leq B^{\ell-1}. \quad (81)$$

Consequently, for any set S of retained predecessors and every t ,

$$\sum_v \mathbf{1}\{t \in J_v\} \sum_{u \in S} \theta_u e^{-\theta_u t} A_{v,u} \leq B^\ell \sup_{u \in S} \theta_u e^{-\theta_u t}, \quad (82)$$

because at most B target windows contain t . Here the supremum is zero when S is empty, and enlarging the actual sum from $u < v$ to all pairs can only increase its left side. The unique rotation cancels the factor $1/\ell$ in the definition of the cycle count. Hence (69) gives

$$\mathbb{E}[\#\{\text{retained } \ell\text{-cycles}\}] \leq \mathbb{E} \sum_{u < v} p_{u,v} A_{v,u},$$

with any further restrictions imposed on the outer pair in both sums.

Suppose first that the terminal depth of v belongs to $\mathcal{D}_{n,j}$, $j \geq J$. The part of $p_{u,v}$ coming from $t < j - h_j$ can be nonzero only if

$$T_{n-r_{n,j}+1-\ell} < j - h_j.$$

Indeed, $v \geq n - r_{n,j} + 1$ and J_v starts at $T_{v-\ell}$. Jensen's inequality and the normalization give

$$S(j - h_j) \geq n e^{-j+h_j} \geq r_{n,j} e^{h_j}.$$

For all large j , a Chernoff lower-tail bound therefore yields

$$\mathbb{P}(T_{n-r_{n,j}+1-\ell} < j - h_j) \leq \exp\{-c r_{n,j} e^{h_j}\}.$$

Pointwise in the common background, the early part of $p_{u,v}$ is at most the indicator of this event, so the total rooted weight for each v is at most $B^{\ell-1}$ times that indicator. Taking expectations and using $|\mathcal{D}_{n,j}| \leq r_{n,j}$, all early-window contributions are bounded by a constant times

$$\sum_{\substack{j \geq J \\ \mathcal{D}_{n,j} \neq \emptyset}} r_{n,j} e^{-c r_{n,j} e^{h_j}} \leq \sum_{r \geq 1} r e^{-c r e^{h_J}} \longrightarrow 0. \quad (83)$$

Here the maxima $r_{n,j}$ of the nonempty shells are distinct integers and $h_j \geq h_J$ for $j \geq J$.

It remains to integrate $p_{u,v}$ over $t \geq j - h_j$. If $u > \beta n$, its terminal depth belongs to a shell $\mathcal{D}_{n,r}$ with $r \geq J_0$. Since $u < v$, one has $r \leq j$, and Lemma 4.5 gives $t \geq j - h_j \geq r - h_r$. By increasing J_0 , we may make the limsup in (55) smaller than $\mathcal{Q}(1)$; then, for all sufficiently large n , every nonempty source shell $r \geq J_0$ satisfies $b_{n,r}(r - h_r) > 1$. As $b_{n,r} \leq \theta_u \leq M$, the map $a \mapsto a e^{-at}$ is decreasing on this range, and hence

$$\theta_u e^{-\theta_u t} \leq b_{n,r} e^{-b_{n,r} t}.$$

Apply (82) separately to each source shell, retaining $u < v$ (and hence $r \leq j$) before enlarging the time range to $[r - h_r, \infty)$, and then sum over the shells. Their contribution is at most

$$B^\ell \sum_{\substack{r \geq J_0 \\ \mathcal{D}_{n,r} \neq \emptyset}} \int_{r-h_r}^{\infty} b_{n,r} e^{-b_{n,r}t} dt = B^\ell \sum_{\substack{r \geq J_0 \\ \mathcal{D}_{n,r} \neq \emptyset}} e^{-b_{n,r}(r-h_r)}, \quad (84)$$

which vanishes as $J_0 \rightarrow \infty$ by (55).

Finally, if $u \leq \beta n$, then $\delta \leq \theta_u \leq M$. For J large, $\delta(J - h_J) > 1$, so on $t \geq j - h_j \geq J - h_J$,

$$\theta_u e^{-\theta_u t} \leq \delta e^{-\delta t}.$$

The same marked-edge bound makes this contribution at most

$$B^\ell \int_{J-h_J}^{\infty} \delta e^{-\delta t} dt = B^\ell e^{-\delta(J-h_J)}, \quad (85)$$

which vanishes as $J \rightarrow \infty$.

Choose first M , then J_0 and the corresponding β , then δ , and finally J , keeping these cutoffs fixed before taking each limsup in n . Equations (83)–(85), followed by the two truncation estimates, prove (80); replacing $1 - e^{-J}$ by an arbitrary $\alpha \uparrow 1$ only requires a rounding check. Integer rounding can add at most one terminal-depth label, which for all large n lies in a shell with index at least $J - 1$. Thus the estimate with $J - 1$ proves the result along $\alpha = 1 - e^{-J}$, and monotonicity gives the general limit. $\square$

Proof of Theorem 1.4. Fix $\alpha < 1$ and nonnegative integers $m_1, \dots, m_L$, and put $r = \sum_{\ell=1}^L \ell m_\ell$. Expanding the joint falling factorial gives an ordered collection of vertex-disjoint cycles. Lemma 5.2 therefore yields

$$\mathbb{E} \prod_{\ell=1}^L (C_{n,\ell}^{(\alpha)})_{m_\ell} \longrightarrow \prod_{\ell=1}^L \lambda_\ell(\alpha)^{m_\ell}, \quad (86)$$

where

$$\lambda_\ell(\alpha) := \frac{1}{\ell} \int_{[0,\alpha]^\ell} \prod_{a=1}^{\ell} \rho(x_a, x_{a+1}) d\mathbf{x}, \quad x_{\ell+1} = x_1.$$

These are the joint factorial moments of independent Poisson variables, so the truncated cycle vector has the asserted limit.

Proposition 5.5 gives the full endpoint tightness (80).

First-moment convergence and (80) show that

$$\sup_{\alpha < \beta < 1} [\lambda_\ell(\beta) - \lambda_\ell(\alpha)] \longrightarrow 0 \quad (\alpha \uparrow 1).$$

Hence the increasing $\lambda_\ell(\alpha)$ have finite limits, necessarily the integrals in (18). Markov's inequality removes the label cutoff in (86) and proves (19). Finally, weak convergence on the countable lattice $\mathbb{Z}_+^L$ gives pointwise convergence of mass functions; Scheffe's lemma upgrades it to total variation convergence. $\square$

6 Power-law endpoint singularities and cycle CLTs

We prove Theorems 1.6 and 1.7 directly under Assumption 1.5. The population estimates needed in the proof are consequences of that assumption, not additional assumptions on a triangular array. The critical pole is treated separately under the hypotheses of Theorem 1.8. The mean-one normalization used in the finite-mean part of the paper is not imposed in this section.

6.1 Population estimates for sampled profiles

Write $E_{n,i} \sim \text{Exp}(\theta_{n,i})$ for the independent clocks and define

$$H_n(t) = \frac{1}{n} \sum_i e^{-t\theta_{n,i}}, \quad G_n(t) = 1 - H_n(t), \quad D_{n,r}(t) = \frac{1}{n} \sum_i \theta_{n,i}^r e^{-t\theta_{n,i}}. \quad (87)$$

We abbreviate $D_{n,1}$ to D_n . Fix a sufficiently small $\epsilon_* > 0$. At each active endpoint, the rates throughout the ϵ_* -neighborhood are between two fixed positive multiples of the specified power. On the complement of the active neighborhoods the rates are in a fixed interval $[d, M] \subset (0, \infty)$. These facts follow by first shrinking the neighborhoods so that their relative errors are less than $1/2$, and then using continuity, positivity, and the finite positive limits at inactive endpoints. No differentiability or monotonicity of f is used.

Lemma 6.1 (Population transforms). *Under Assumption 1.5, at an active right endpoint, put $A_- = \Gamma(1 + 1/\beta)c_R^{-1/\beta}$. For some $\zeta > 0$, uniformly as $t \rightarrow \infty$ and $nt^{-1/\beta} \rightarrow \infty$,*

$$\begin{aligned} H_n(t) &= A_- t^{-1/\beta} \left(1 + O\left(t^{-\zeta} + (nt^{-1/\beta})^{-1}\right) \right), \\ D_n(t) &= \frac{A_-}{\beta} t^{-1-1/\beta} \left(1 + O\left(t^{-\zeta} + (nt^{-1/\beta})^{-1}\right) \right). \end{aligned} \quad (88)$$

At an active left endpoint, put $A_+ = \Gamma(1 - 1/\alpha)c_L^{1/\alpha}$. Uniformly as $t \downarrow 0$ and $nt^{1/\alpha} \rightarrow \infty$,

$$\begin{aligned} G_n(t) &= A_+ t^{1/\alpha} \left(1 + O\left(t^\zeta + (nt^{1/\alpha})^{-1}\right) \right), \\ D_n(t) &= \frac{A_+}{\alpha} t^{1/\alpha-1} \left(1 + O\left(t^\zeta + (nt^{1/\alpha})^{-1}\right) \right). \end{aligned} \quad (89)$$

The constants and exponent may depend on the active endpoint. After deleting any fixed number r of clocks, the same statements hold for the sums divided by n , with constants uniform in the deleted labels.

Proof. We estimate the unweighted transforms together with $tD_n(t)$. For the explicit monotone or unimodal kernels below, midpoint averaging introduces an error bounded by their supremum times $O(n^{-1})$.

At the right endpoint, reflect the sampling grid and compare $f(1-s)$ with cs^β , where $c = c_R$. The pure-power kernels have integrals

$$\int_0^\infty e^{-cts^\beta} ds = A_- t^{-1/\beta}, \quad \int_0^\infty cts^\beta e^{-cts^\beta} ds = \frac{A_-}{\beta} t^{-1/\beta}.$$

Their suprema are uniformly bounded, and their tails beyond $s = 1$ are exponentially small for $t \geq 1$.

The endpoint expansion and the mean value theorem bound the errors in both kernels by

$$Cts^{\beta+\eta}e^{-dts^\beta}$$

near zero, after absorbing a linear factor into the exponential. This unimodal envelope has integral $O(t^{-(1+\eta)/\beta})$ and supremum $O(t^{-\eta/\beta})$. Away from the endpoint, both rates are bounded below, so both kernels are exponentially small; here $t\alpha e^{-ta} \leq (2/e)e^{-ta/2}$. Consequently,

$$|H_n(t) - A_- t^{-1/\beta}| + \left| tD_n(t) - \frac{A_-}{\beta} t^{-1/\beta} \right| \leq C \left(t^{-(1+\eta)/\beta} + n^{-1} \right).$$

Division by the leading terms proves (88).

At the left endpoint, choose $0 < \eta' < \min\{\eta, \alpha - 1\}$ and put $q = \alpha - \eta' > 1$. With $c = c_L$, the pure-power integrals are

$$\int_0^\infty (1 - e^{-cts^{-\alpha}}) ds = A_+ t^{1/\alpha}, \quad \int_0^\infty cts^{-\alpha} e^{-cts^{-\alpha}} ds = \frac{A_+}{\alpha} t^{1/\alpha}.$$

Their midpoint errors are $O(n^{-1})$, and their tails beyond 1 are $O(t)$. The perturbation errors near zero are bounded by

$$Cts^{-q}e^{-dts^{-q}} \leq Cts^{-q}e^{-dts^{-q}}, \quad 0 < s < 1.$$

The last kernel has integral $O(t^{1/q})$ and bounded supremum. Away from zero, boundedness of both rates gives an $O(t)$ contribution. Since $t \leq t^{1/q}$ for $0 < t \leq 1$,

$$|G_n(t) - A_+ t^{1/\alpha}| + \left| tD_n(t) - \frac{A_+}{\alpha} t^{1/\alpha} \right| \leq C(t^{1/q} + n^{-1}).$$

This gives (89) with $\zeta = 1/q - 1/\alpha > 0$.

Finally, deleting at most r terms changes either unweighted sum divided by n by at most r/n , and changes tD_n by at most $r/(en)$. These errors are already allowed, uniformly in the deleted labels. $\square$

Lemma 6.2 (Quantiles and deterministic weight bounds). *At an active right endpoint define $t_{n,m}^R$ by $H_n(t_{n,m}^R) = m/n$; at an active left endpoint define $t_{n,m}^L$ by $G_n(t_{n,m}^L) = m/n$. There is $\xi > 0$ such that, uniformly for $m \rightarrow \infty$ and $m/n \rightarrow 0$,*

$$\begin{aligned} t_{n,m}^R &= (A_- n/m)^\beta (1 + O(e_{n,m})), & \theta_{n,n-m+1} t_{n,m}^R &= q_R (1 + O(e_{n,m})), \\ \frac{\theta_{n,n-m+1}}{nD_n(t_{n,m}^R)} &= \frac{\beta q_R}{m} (1 + O(e_{n,m})), \end{aligned} \tag{90}$$

$$\begin{aligned} t_{n,m}^L &= (m/(A_+ n))^\alpha (1 + O(e_{n,m})), & \theta_{n,m} t_{n,m}^L &= q_L (1 + O(e_{n,m})), \\ \frac{\theta_{n,m}}{nD_n(t_{n,m}^L)} &= \frac{\alpha q_L}{m} (1 + O(e_{n,m})), \end{aligned} \tag{91}$$

where $e_{n,m} = (m/n)^\xi + m^{-1}$ and q_R, q_L are exactly (21)–(22). For $t = t_{n,m}^\sigma$ and $t/4 \leq s \leq 4t$,

$$nD_n(s) \asymp m/t, \quad nD_{n,2}(s) \leq Cm/t^2. \tag{92}$$

The corresponding survivor mean nH_n on the right, or arrival mean nG_n on the left, changes by at least cum between t and $(1 \pm u)t$, for $0 < u \leq 1/2$. Moreover $D_n((1 \pm u)t)/D_n(t) = 1 + O(u)$.

After deleting at most r clocks, the total rate in a gap with h survivors at the right endpoint is deterministically at least

$$cn^{-\beta}h^{\beta+1}. \quad (93)$$

At a gap after $h + O(r)$ early arrivals at the left endpoint it is at least

$$c_r n^\alpha h^{1-\alpha}, \quad (94)$$

provided $h \geq 4r + 4$ and $h \leq \epsilon_* n/4$. For ranks in any fixed compact subinterval of $(0, 1)$, the deterministic quantiles lie in a compact subinterval of $(0, \infty)$, nD_n is comparable to n , and the remaining rate is deterministically at least cn , uniformly after bounded deletions.

Proof. Put $x = m/n$. Positivity of the rates makes H_n strictly decreasing and G_n strictly increasing, so the quantiles exist uniquely for $0 < m < n$.

We first justify inversion in the joint regime. On the right, choose a fixed large S where the power error in Lemma 6.1 is a small fraction of the leading term. For x small and m large, $H_n(S) > x$, hence $t = t_{n,m}^R > S$. At this quantile the additive estimate gives

$$|x - A_- t^{-1/\beta}| \leq C t^{-\eta/\beta} A_- t^{-1/\beta} + C/n.$$

Absorbing the small errors first gives $t \asymp x^{-\beta}$; substitution back into the inequality gives the stated relative expansion. The same argument on the left, starting at a fixed small S with $G_n(S) > x$, gives $t_{n,m}^L \asymp x^\alpha$ and then its relative expansion. Thus the population remainders become positive powers of x .

The sampled endpoint expansions have an additional $O(m^{-1})$ error because the endpoint distance is $x(1 - 1/(2m))$. Multiplying by the quantile expansions gives the rate-time products, using

$$c_R A_-^\beta = q_R, \quad c_L A_+^{-\alpha} = q_L.$$

The weighted population estimates give

$$nt_{n,m}^R D_n(t_{n,m}^R) = \frac{m}{\beta}(1 + O(e_{n,m})), \quad nt_{n,m}^L D_n(t_{n,m}^L) = \frac{m}{\alpha}(1 + O(e_{n,m})).$$

Choose a common positive ξ below all the resulting error exponents. Division then proves the formulas for $\theta/(nD_n)$.

For $t = t_{n,m}^\sigma$, Lemma 6.1 applies uniformly on $[t/8, 8t]$ and gives $nD_n(s) \asymp m/t$. The inequality

$$D_{n,2}(s) \leq \frac{2}{es} D_n(s/2)$$

therefore proves the second-moment bound on $[t/4, 4t]$. Integrating the exact derivatives

$$H'_n = -D_n, \quad G'_n = D_n, \quad D'_n = -D_{n,2}$$

gives the mean separation and relative weight stability. No asymptotic formula is differentiated.

At the right endpoint, the global bound

$$\theta_{n,i} \geq c((n - i + 1)/n)^\beta$$

implies that any h remaining labels have total rate at least

$$cn^{-\beta} \sum_{a=1}^h a^\beta \geq c' n^{-\beta} h^{\beta+1}.$$

At the left endpoint, after at most $2h$ total removals, at least h of the first $3h$ labels remain. Each has rate at least $c(n/h)^\alpha$, proving the left floor. Bounded deletion and displacement budgets are absorbed into this count by increasing the fixed block multiple if necessary and shrinking the endpoint neighborhood.

Finally, the population estimates, or global rate bounds at inactive endpoints, provide fixed $0 < a < b$ with $G_n(a) < \delta$ and $H_n(b) < \delta$. Thus interior quantiles lie in $[a, b]$. A fixed interior block of bounded positive rates gives $D_n \geq c$ there, while $D_n(s) \leq 1/(es)$ gives the upper bound. Bounded deletions preserve these estimates. A linear number of survivors has total rate at least cn , by the right floor when that endpoint is active and by a global positive rate lower bound otherwise. $\square$

6.2 Joint insertion estimates

Throughout this subsection we assume Assumption 1.5. The number of marked labels is fixed; constants may depend on this number and on a fixed moment order. No monotonicity of the sampled profile is used. Write

$$\iota_R(m) = n - m + 1, \quad \iota_L(m) = m, \quad d_R(i) = n - i + 1, \quad d_L(i) = i.$$

Choose a sufficiently small fixed $\delta > 0$. At an active corner let

$$x_R(a, h) = (a/h)^\beta, \quad x_L(a, h) = (h/a)^\alpha, \quad K_\sigma(a, h) = h^{-1}k_\sigma(a/h),$$

where

$$k_R(u) = \beta q_R u^\beta e^{-q_R u^\beta}, \quad k_L(u) = \alpha q_L u^{-\alpha} e^{-q_L u^{-\alpha}}.$$

Here q_R, q_L have the values in (21)–(22). For a sufficiently small $c > 0$ put

$$H_R(a, h) = \frac{1}{h}(a/h)^\beta e^{-c(a/h)^\beta}, \quad H_L(a, h) = \frac{1}{h}(h/a)^\alpha e^{-c(h/a)^\alpha}.$$

The constant c will always retain exponential slack: decreasing it a fixed number of times is allowed. In particular, any fixed polynomial in x_σ can be absorbed by this slack.

The exact marked probability space. Delete a set D of at most r labels. Order the remaining independent clocks as

$$0 = T_0^D < T_1^D < \cdots < T_N^D < T_{N+1}^D = \infty, \quad N = n - |D|.$$

For a label $i \in D$ define its conditional probability of entering gap q by

$$X_{i,q}^D = \int_{T_q^D}^{T_{q+1}^D} \theta_{n,i} e^{-\theta_{n,i}t} dt. \quad (95)$$

Suppose that distinct marked labels $i_1, \dots, i_s$ are required to have distinct full ranks $j_1, \dots, j_s$, and take $D = \{i_1, \dots, i_s\}$. If $j_{e_1} < \cdots < j_{e_s}$, set

$$q_{e_b} = j_{e_b} - b, \quad 1 \leq b \leq s. \quad (96)$$

For each repeated value q , integrate the corresponding marked clocks over the ordered simplex in (T_q^D, T_{q+1}^D) , in the order of their demanded ranks. Denote this integral by Z_q^D . Independence of the marked clocks gives the exact identity

$$\mathbb{P}(R_{n,i_e} = j_e \text{ for all } e) = \mathbb{E} \prod_q Z_q^D, \quad \prod_q Z_q^D \leq \prod_e X_{i_e, q_e}^D. \quad (97)$$

The second assertion follows by enlarging each ordered simplex to its product box. If the q_e are distinct, it is an equality. This accounts for adjacent demanded ranks without assigning independent ranks to clocks which must occupy one ordered gap.

Lemma 6.3 (Moments of one insertion). *Fix positive integers r, p_0 . There are $h_0, \nu_0, \nu, c, C > 0$ such that the following assertions hold, uniformly after deletion of at most r labels and uniformly for gap indices within $r + 1$ of the full-rank gap under consideration. At an active corner, for $h_0 \leq a, h \leq \delta n$ and $1 \leq p \leq p_0$, put*

$$\mathcal{M}_\sigma(a, h) = \{x_\sigma(a, h) \leq (a \wedge h)^{\nu_0}\}, \quad E_R(a, h) = e^{-ca^\nu} \mathbf{1}_{\mathcal{M}_R(a, h)^c}, \quad E_L(a, h) = e^{-ch^\nu} \mathbf{1}_{\mathcal{M}_L(a, h)^c}.$$

Then

$$\|X_{\iota_\sigma(a), q}^D\|_p \leq C\{H_\sigma(a, h) + E_\sigma(a, h)\}. \quad (98)$$

The exceptional kernels have bounded row and column sums. More precisely, after decreasing ν and c ,

$$\sup_{a \geq A} \sum_{h \geq A} E_\sigma(a, h) + \sup_{h \geq A} \sum_{a \geq A} E_\sigma(a, h) \leq C e^{-cA^\nu}. \quad (99)$$

The same restricted-sum assertion holds for $H_\sigma \mathbf{1}_{\mathcal{M}_\sigma^c}$.

Proof. Fix $0 < \nu_0 \leq 1$, increase h_0 to absorb the deletion and gap-shift budgets, and shrink δ as necessary. Write

$$i = \iota_\sigma(a), \quad \theta = \theta_{n,i}, \quad s = T_q^D, \quad x = x_\sigma(a, h).$$

Conditional on the background elimination chain, the nonfinal gap length is $\Delta = \xi/W$, where ξ is standard exponential and independent of the preceding spacings, hence of s . Lemma 6.2 and endpoint comparability give

$$W \geq cn^{-\beta} h^{\beta+1} (R), \quad W \geq cn^\alpha h^{1-\alpha} (L), \quad \theta/W \leq Cx/h.$$

Since $X_{i,q}^D = e^{-\theta s}(1 - e^{-\theta \Delta})$, integration over ξ yields

$$\mathbb{E}[(X_{i,q}^D)^p] \leq p!(Cx/h)^p \mathbb{E}e^{-p\theta s}. \quad (100)$$

Put $\tau_h^R = (n/h)^\beta$ and $\tau_h^L = (h/n)^\alpha$. At a sufficiently small fixed multiple of τ_h^σ , the population estimates place the survivor mean on the right well above the gap threshold, or the arrival mean on the left well below it. These comparisons retain a fixed margin after the allowed deletions and shifts. Chernoff bounds therefore give

$$\mathbb{P}(s < k\tau_h^\sigma) \leq Ce^{-dh}, \quad \mathbb{E}e^{-p\theta s} \leq C(e^{-dx} + e^{-dh}), \quad (101)$$

using $\theta\tau_h^\sigma \asymp x$. On the moderate set, $x \leq h$, so (100), followed by taking p -th roots, gives $C(x/h)e^{-cx}$.

For the right extreme region, put $\gamma = \beta/(\beta + 1)$. If $h \leq a^\gamma$, repeat the survivor-count argument at comparison depth $y = a^\gamma$, that is, at time $k(n/y)^\beta$. The mean is a sufficiently large multiple of y , and $\theta(n/y)^\beta \asymp (a/y)^\beta = y$. Thus the survival expectation is at most Ce^{-da^γ} . If $h > a^\gamma$, use (101). Together,

$$\mathbb{E}e^{-p\theta s} \leq C(e^{-dx} + e^{-da^\gamma}). \quad (102)$$

On $\mathcal{M}_R^c$,

$$a > h, \quad x > a^{\beta\nu_0/(\beta+\nu_0)}, \quad x/h \leq a^\beta.$$

Choosing $\nu < \min\{\beta\nu_0/(\beta + \nu_0), \gamma\}$, polynomial absorption in (100) gives Ce^{-ca^ν} .

Similarly, on $\mathcal{M}_L^c$,

$$h > a, \quad x > h^{\alpha\nu_0/(\alpha+\nu_0)}, \quad x/h \leq h^\alpha.$$

Equations (100) and (101) give Ce^{-ch^ν} for $\nu < \min\{\alpha\nu_0/(\alpha + \nu_0), 1\}$. Constants can be chosen uniformly for $1 \leq p \leq p_0$, proving the insertion bound.

For the restricted sums, keep these kernels fixed. The right exceptional support satisfies $h < a^{\beta/(\beta+\nu_0)} < a$. Its row sum is therefore at most ae^{-ca^ν} , and its restricted column sum is at most $\sum_{a \geq A} e^{-ca^\nu}$. Both give the required stretched-exponential tail after weakening the tail constant. The left case follows with the roles interchanged.

Finally, on the extreme sets, the displayed lower bounds for x and the inequality $xe^{-cx} \leq Ce^{-cx/2}$ dominate the restricted typical kernels by exceptional kernels with positive stretched-exponential exponents. The same summation argument applies. Taking common smaller constants and exponents in the tail bounds completes the proof. $\square$

Lemma 6.4 (Global domination and excursions). *For every fixed r there is a deterministic nonnegative matrix $M_n(i, j)$ such that every cylinder of at most r distinct marked labels and distinct demanded ranks satisfies*

$$\mathbb{P}(R_{n, i_e} = j_e \text{ for all } e) \leq C_r \prod_e M_n(i_e, j_e). \quad (103)$$

It can be chosen to have

$$\sup_i \sum_j M_n(i, j) \leq C_r, \quad (104)$$

and, with $m = d_\sigma(j)$ at an active corner,

$$\sup_i M_n(i, j) \leq C_r/m \quad (m \leq \delta n), \quad \sup_i M_n(i, j) \leq C_r/n \quad (\delta n \leq j \leq (1 - \delta)n). \quad (105)$$

Its column sums are bounded for active-corner and middle targets. For source and target in the same active corner it is bounded by $C_r(H_\sigma + E_\sigma)$, apart from a fixed number of endpoint indices, which obey the bounds in the proof below. For some $\kappa > 0$ it also has

$$\sup_i \sum_j M_n(i, j) \left(\frac{d_R(j)}{d_R(i)} \right)^\kappa \leq C_r \quad \text{if } R \text{ is active}, \quad (106)$$

$$\sup_i \sum_j M_n(i, j) \left(\frac{d_L(i)}{d_L(j)} \right)^\kappa \leq C_r \quad \text{if } L \text{ is active}. \quad (107)$$

In particular, for fixed ℓ , the expected number of ℓ -cycles whose largest vertex is $\iota_\sigma(m)$ is at most C_ℓ/m at an active corner and at most C_ℓ/n in the middle. Among cycles rooted at active-corner distances $A \leq m \leq B \leq \delta n$, requiring every vertex to have its distance in $[A, B]$ discards only $O_\ell(1)$ in expectation.

Proof. Put $p = r + 1$. Define $M_n(i, j)$ as the maximum of $\|X_{i,q}^D\|_p$ over deletion sets of size at most p and valid background gaps satisfying $|q - (j - 1)| \leq p + 1$. Here the gap integral defines $X_{i,q}^D$ for every source label, whether or not it belongs to D . This is a finite maximum, and $0 \leq M_n(i, j) \leq 1$. All estimates below are uniform in these choices, which may vary from entry to entry.

For a cylinder with $s \leq r$ marks, the ordered-simplex bound and Hölder's inequality give

$$\mathbb{P}(R_{n,i_e} = j_e \text{ for all } e) \leq \mathbb{E} \prod_e X_{i_e, q_e}^D \leq \prod_e \|X_{i_e, q_e}^D\|_s \leq \prod_e M_n(i_e, j_e).$$

Thus it remains to establish the matrix estimates.

For targets away from a fixed number of endpoint indices, the moment reduction in Lemma 6.3 gives the ordinary bound

$$\frac{C}{d_j} \theta_{n,i} \tau_j e^{-c\theta_{n,i} \tau_j}, \quad (d_j, \tau_j) = \begin{cases} (j, (j/n)^\alpha), & \text{active left targets,} \\ (n - j + 1, (n/(n - j + 1))^\beta), & \text{active right targets,} \\ (n, 1), & \text{middle targets.} \end{cases} \quad (108)$$

For sources and targets in the same active corner, Lemma 6.3 supplies the exceptional term CE_σ . Outside the right source block, the rates are bounded below; its alternative-time argument, now at comparison depth $n^{\beta/(\beta+1)}$, supplies an exceptional term Ce^{-cn^ν} . Outside the left source block the rates are bounded above, so $\theta_i(j/n)^\alpha < 1$ after shrinking the target block and the moderate estimate suffices. Middle targets have remainder Ce^{-cn} . Polynomial rate prefactors are absorbed into these exponentials.

For the finitely many earliest active-left targets, the remaining-rate floor gives $C\theta_i/n^\alpha$. For the finitely many active-right terminal targets, including the infinite final gap, bound the insertion factor by survival to an earlier fixed terminal depth. The alternative-time estimate gives Ce^{-ca^ν} for right-source depth a , and Ce^{-cn^ν} outside that block.

At an inactive left endpoint, rates are bounded above and the remaining rate is at least cn ; retain the bound $C\theta_i/n$. At an inactive right endpoint, rates are bounded below, and a linear number are also bounded above by a fixed M . At time $c_0 \log(n/m)$, this latter group has at least $c\sqrt{nm}$ expected survivors when $c_0 M < 1/2$. Chernoff and the moment reduction give

$$M_n(i, j) \leq \frac{C\theta_i}{m} (m/n)^{c\theta_i} + Ce^{-c\sqrt{nm}}, \quad m = n - j + 1. \quad (109)$$

For bounded m , the survival factor alone gives the same bound after reducing c . These estimates cover all gaps.

The ordinary row sums are bounded by sum-integral comparison, using $u = \theta_i \tau_j$. The active-corner integrals reduce to constant multiples of $\int_0^\infty e^{-cu} du$; the inactive terminal integral is

$$\int_0^1 \theta_i u^{c\theta_i-1} du = 1/c.$$

The middle and inactive-left row sums are bounded directly. Exceptional sums are bounded by Lemma 6.3 and stretched-exponential decay.

For active columns, the same comparison gives

$$\sum_a (h/a)^\alpha e^{-c(h/a)^\alpha} \leq Ch, \quad \sum_a (a/h)^\beta e^{-c(a/h)^\beta} \leq Ch.$$

Sources outside the corresponding corner contribute at most $Cn(h/n)^\alpha \leq Ch$ on the left and $Cne^{-c(n/h)^\beta} \leq Ch$ on the right. Division by the target depth proves the column bounds. Middle columns follow from boundedness of $\theta e^{-c\theta}$. The fixed endpoint columns use $\sum_a a^{-\alpha} < \infty$ and $\sum_a e^{-ca^\nu} < \infty$. Finally, boundedness of ue^{-cu} , together with the exceptional support restrictions, gives the target maxima. Enlarging constants handles the finitely many small array sizes.

Choose a common $\kappa > 0$ with $\kappa < \beta$ at an active right endpoint and $\kappa < \alpha$ at an active left endpoint. The local ordinary row integrals remain bounded after inserting the respective weights $(h/a)^\kappa$ and $(a/h)^\kappa$. Local exceptional edges have weight at most one.

For a right source of depth a , the total mass outside the right target block is $O(\theta_i) = O((a/n)^\beta)$, which absorbs $(n/a)^\kappa$. Sources outside that source block have bounded right-depth ratios, so their ordinary row bound suffices. For the left weight, targets outside the left block have bounded weight. Sources outside the left source block have bounded rates, and their left-target contribution is bounded by

$$C\theta_i n^{-\alpha} \sum_{h \leq \delta n} h^{\alpha-1} (n/h)^\kappa \leq C.$$

The fixed endpoint estimates above also satisfy these weighted bounds. This proves both global weighted row estimates.

For cycles of length ℓ , use the construction with a sufficient mark budget. At a root v , bound the closing edge by $C/d_\sigma(v)$, or C/n in the middle, and sum the remaining edges using the row bound. Dropping distinctness and maximality only enlarges the sum, proving the rooted-cycle estimates.

Finally, a largest-label right root of depth m can leave $[A, B]$ only through depths greater than B ; a left root can leave only through depths below A . Iterating the weighted row bounds along an open path, and summing over its possible escape positions, gives respective path bounds

$$C_\ell(m/B)^\kappa, \quad C_\ell(A/m)^\kappa.$$

Multiplying by the closing-edge bound C_ℓ/m and summing over $A \leq m \leq B$ gives $O_\ell(1)$ in either case. The row bounds are global, so these estimates include paths through the middle or the opposite corner. $\square$

Order-time concentration with a marked background. Let t_h^σ denote the deterministic full-array mean quantile: $nH_n(t_h^R) = h$ or $nG_n(t_h^L) = h$, and let $\mu_h^\sigma = nD_n(t_h^\sigma)$. These equations also specify the notation when a bounded deletion is made; the deletion is treated as an error, not as a change of quantile. Lemma 6.2 and Chernoff give, at a gap of corner distance $h + O_r(1)$,

$$\mathbb{P}\left(|T_q^D/t_h^\sigma - 1| > Ch^{-1/4} \text{ or } |W_q^D/\mu_h^\sigma - 1| > Ch^{-1/4}\right) \leq Ce^{-c\sqrt{h}}. \quad (110)$$

Here is the weight argument explicitly. For independent Bernoulli variables B_i with means $p_i = e^{-\theta_i s}$, the inequalities $\log(1+z) \leq z$ and $e^v - 1 - v \leq \frac{1}{2}v^2 e^{|v|}$ give

$$\log \mathbb{E} \exp \left\{ \lambda \sum_i \theta_i (B_i - p_i) \right\} \leq \frac{\lambda^2}{2} \sum_i \theta_i^2 e^{-\theta_i(s-|\lambda|)} \leq \frac{\lambda^2}{2} nD_{n,2}(s/2), \quad |\lambda| \leq s/2.$$

Here $D_{n,2}(s) = n^{-1} \sum_i \theta_i^2 e^{-s\theta_i}$. At $s = t_\pm = (1 \pm h^{-1/4})t_h^\sigma$, the deterministic bound is $nD_{n,2}(s/2) \leq Ch/(t_h^\sigma)^2$. Taking λ equal to a sufficiently small constant times $h^{-1/4}t_h^\sigma$, with

either sign, proves an exponential bound $2e^{-c\sqrt{h}}$ at deviation $h^{3/4}/t_h^\sigma$. Deleting a fixed number of clocks changes a weighted mean by at most C_r/t_h^σ , since $\theta e^{-\theta t} \leq C/t$. The count-mean separation places T_q^D between t_- and t_+ with error $Ce^{-c\sqrt{h}}$; monotonicity of the surviving weight in time and the deterministic mean stability then prove the display. Monotonicity of the profile is not involved.

Proposition 6.5 (Joint local insertion law). *Fix r . For each of at most r marked edges choose an active corner σ_e , a source $i_e = \iota_{\sigma_e}(a_e)$ and a demanded full rank $j_e = \iota_{\sigma_e}(h_e)$, with distinct sources and distinct ranks. Different edges may belong to different active corners. Suppose that $h_0 \leq a_e, h_e \leq \delta n$, that all edges are moderate in the sense of Lemma 6.3, and that $|j_e - j_f| > 2r$ for $e \neq f$. There are $\kappa, C > 0$, depending on r , such that*

$$\left| \mathbb{P}(R_{n,i_e} = j_e \text{ for all } e) - \prod_e K_{\sigma_e}(a_e, h_e) \right| \leq C \prod_e H_{\sigma_e}(a_e, h_e) \sum_e \left\{ (a_e \wedge h_e)^{-\kappa} + ((a_e \vee h_e)/n)^\kappa \right\}. \quad (111)$$

For all other configurations the product domination in (103) remains valid, including adjacent demanded ranks. In particular, if every endpoint distance belongs to $[A, B]$, the error on the moderate separated configurations is bounded by $C(A^{-\kappa} + (B/n)^\kappa) \prod_e H_{\sigma_e}(a_e, h_e)$.

Proof. Delete the marked labels and order the demanded ranks as in (96). Their separation makes the corresponding background gaps distinct and, after increasing h_0 , nonfinal. Write

$$X_e = X_{i_e, q_e}^D, \quad \xi_e = W_e(T_{q_e+1}^D - T_{q_e}^D), \quad K_e = K_{\sigma_e}(a_e, h_e), \quad H_e = H_{\sigma_e}(a_e, h_e).$$

The exact insertion identity gives

$$\mathbb{P}(R_{n,i_e} = j_e \text{ for all } e) = \mathbb{E} \prod_e X_e.$$

Conditional on the background elimination chain, the selected normalized gaps are independent standard exponentials. Consequently, with s the number of marks and $Y = \prod_e \xi_e$,

$$\mathbb{E}Y = 1, \quad \mathbb{E}(\xi_e Y) = 2, \quad \mathbb{E}Y^2 = 2^s.$$

We use these identities without asserting independence of the insertion factors.

The quantile and remaining-weight estimates give measurable events G_e with

$$\mathbb{P}(G_e^c) \leq Ch_e^{-1/2},$$

on which the gap starting time and remaining rate differ relatively from their deterministic values by $O(h_e^{-1/4})$. Together with the sampled endpoint expansions, these estimates imply, for some $\rho > 0$,

$$K_e \leq CH_e, \quad X_e \leq CH_e \xi_e, \quad |X_e - K_e \xi_e| \leq CH_e \left(\varepsilon_e \xi_e + \frac{\xi_e^2}{h_e} \right),$$

where

$$\varepsilon_e = h_e^{-1/4} + (a_e \wedge h_e)^{-1} + \left(\frac{a_e \vee h_e}{n} \right)^\rho.$$

Indeed, expand $1 - e^{-\theta_e \Delta_e}$ to first order, using $\Delta_e = \xi_e/W_e$, and compare the coefficient $e^{-\theta_e T_{qe}^D} \theta_e/W_e$ with K_e . The exponential slack in H_e absorbs the resulting powers of $x_{\sigma_e}(a_e, h_e)$. All estimates are uniform in the bounded deletions and rank displacements.

Put $G = \bigcap_e G_e$. Telescoping the product on G gives

$$\left| \prod_e X_e - \left(\prod_e K_e \right) Y \right| \leq C_r \left(\prod_e H_e \right) Y \sum_e \left(\varepsilon_e + \frac{\xi_e}{h_e} \right).$$

Integration and the displayed gap moments bound this contribution by $C_r \prod_e H_e \sum_e (\varepsilon_e + h_e^{-1})$.

On G^c , Hölder and the moderate-case estimate of Lemma 6.3 give

$$\mathbb{E} \left[\mathbf{1}_{G^c} \prod_e X_e \right] \leq \mathbb{P}(G^c)^{1/2} \prod_e \|X_e\|_{2s} \leq C_r \left(\prod_e H_e \right) \mathbb{P}(G^c)^{1/2}. \quad (112)$$

The comparison product has the same bound by Cauchy–Schwarz, since $\mathbb{E}Y^2 = 2^s$ and $K_e \leq CH_e$. Moreover,

$$\mathbb{P}(G^c)^{1/2} \leq C_r \left(\sum_e h_e^{-1/2} \right)^{1/2} \leq C_r \sum_e h_e^{-1/4}.$$

Thus no independence between G and the selected gaps is needed.

Choose a common $0 < \kappa \leq \min\{\rho, 1/8\}$, and take the moderation exponent to be $1/16$. Every error above is bounded by a constant times

$$\sum_e \left\{ (a_e \wedge h_e)^{-\kappa} + \left(\frac{a_e \vee h_e}{n} \right)^\kappa \right\}.$$

Since $\mathbb{E}Y = 1$, this proves (111). Restricting all distances to $[A, B]$ gives the interval estimate, absorbing the number of marks into the constant. Global domination for the remaining configurations follows from Lemma 6.4. The empty configuration is immediate. $\square$

Lemma 6.6 (An integrable density envelope). *Let $E_1, \dots, E_n$ be independent continuous random variables with densities $g_1, \dots, g_n$, and let R_i be the rank of E_i in increasing order. Suppose $S \subseteq [n]$ and $h : \mathbb{R} \rightarrow [0, \infty)$ satisfy $g_i(t) \leq h(t)$ for $i \in S$ and $\int_{\mathbb{R}} h(t) dt < \infty$. For every fixed $\ell \geq 1$, there is a constant K_ℓ , independent of n , the clock laws, and S , such that*

$$\mathbb{E} \# \{i \in S : i \text{ belongs to an } \ell\text{-cycle of } R\} \leq K_\ell \int_{\mathbb{R}} h(t) dt.$$

Proof. We prove the assertion with $K_\ell = 3^\ell$. For a background vector $x = (x_1, \dots, x_n)$, let $x^{i,t}$ be obtained by replacing x_i with t , and write

$$F_i(x) = \mathbf{1}\{i \text{ belongs to an } \ell\text{-cycle of the rank permutation of } x\}.$$

Independence and integration in the i -th coordinate give

$$\mathbb{E}F_i(E) = \int_{\mathbb{R}} g_i(t) \mathbb{E}F_i(E^{i,t}) dt.$$

Thus density domination and Tonelli's theorem yield

$$\mathbb{E} \sum_{i \in S} F_i(E) \leq \int_{\mathbb{R}} h(t) \mathbb{E} \sum_{i \in S} F_i(E^{i,t}) dt.$$

It remains to bound the sum inside the expectation. Fix distinct background values $x_1, \dots, x_n$ and a time t different from all of them. Let R be the background rank permutation and $N(t) = \#\{j : x_j < t\}$. Replacing one coordinate changes the rank of every unchanged clock by at most one; the replacement clock itself has rank within one of $N(t) + 1$.

Suppose i belongs to an ℓ -cycle of the modified rank permutation $R^{i,t}$. Following that cycle from the successor of i back to i gives a sequence

$$v_1 = R^{i,t}(i), \quad v_2, \dots, v_\ell = i$$

satisfying

$$|v_1 - (N(t) + 1)| \leq 1, \quad |v_{a+1} - R(v_a)| \leq 1 \quad (1 \leq a < \ell).$$

Indeed, exact cycle length ensures that the intermediate sources v_a differ from the replaced label i . There are at most three choices for v_1 , and at most three for each subsequent label. Hence there are at most 3^ℓ such sequences. Their final labels determine their roots, so

$$\sum_{i \in S} F_i(x^{i,t}) \leq 3^\ell.$$

For each fixed t , independence and the existence of densities ensure that the background values are almost surely distinct and avoid t . Applying this deterministic bound in the preceding integral proves

$$\mathbb{E} \#\{i \in S : i \text{ belongs to an } \ell\text{-cycle of } R\} \leq 3^\ell \int_{\mathbb{R}} h(t) dt.$$

□

6.3 Cycle bounds and logarithmic traces

Throughout this subsection and the next two we assume Assumption 1.5. Constants may depend on the profile and on a fixed upper bound for the number of marked clocks. Write

$$d_R(i) = n + 1 - i, \quad d_L(i) = i.$$

A cycle is always rooted at its largest label, as in the introduction. Its root therefore has the smallest right distance and the largest left distance among its vertices.

We first record consequences of the insertion estimates that hold before any asymptotic approximation is made. The domination matrix in Lemma 6.4 can be chosen for any fixed number of marked clocks. We denote it by F ; thus the probability of a prescribed finite collection of assignments is bounded by the product of its F entries. Its row sums are bounded. For an active target corner σ , its entries satisfy

$$F(i, j) \leq \frac{C}{d_\sigma(j)} \quad \text{when } j \text{ belongs to a sufficiently small fixed neighborhood of that corner.} \quad (113)$$

The same lemma supplies the weighted row bounds, with some $\kappa > 0$,

$$\sum_j F(i, j) \left(\frac{d_R(j)}{d_R(i)} \right)^\kappa \leq C, \quad \text{at an active right corner,} \quad (114)$$

$$\sum_j F(i, j) \left(\frac{d_L(i)}{d_L(j)} \right)^\kappa \leq C, \quad \text{at an active left corner.} \quad (115)$$

Here and below the global versions of these bounds include paths that leave the endpoint neighborhood. Products along any fixed number of steps have the same bounds with a different constant, because the weight in each display telescopes along the path.

Lemma 6.7 (Root and excursion bounds). *Fix L . At an active corner σ , the expected number of ℓ -cycles with root distance m is at most C_L/m , uniformly for $\ell \leq L$ and m in a sufficiently small fixed endpoint neighborhood. The contribution of a fixed middle interval, and that of every inactive endpoint, is bounded uniformly in n .*

For $1 \leq A \leq B$ within an active endpoint neighborhood, the expected number of cycles whose root distance lies in $[A, B]$ but which have a vertex outside that distance interval is $O_L(1)$. More generally, for $R \geq 1$, the expected number with root distance in $[A, B]$ and with some vertex at logarithmic distance at least $\log R$ from the root is at most

$$C_L(1 + \log(B/A))R^{-\kappa_L}. \quad (116)$$

Proof. The active-corner root bound and the $O_\ell(1)$ interval-exit bound follow from Lemma 6.4. Taking maxima of the constants and a common smaller endpoint neighborhood makes both estimates uniform for $1 \leq \ell \leq L$.

In a fixed middle interval or a sufficiently small neighborhood of an inactive endpoint, the sampled rates lie in some fixed interval $[d, M] \subset (0, \infty)$. Their densities are dominated by

$$h(t) = Me^{-dt}\mathbf{1}_{\{t \geq 0\}}, \quad \int_{\mathbb{R}} h(t) dt = M/d.$$

Lemma 6.6 therefore bounds the expected number of vertices in these regions belonging to an ℓ -cycle by $3^\ell M/d$. This also bounds the number of cycles rooted there, uniformly for $\ell \leq L$.

It remains to prove the logarithmic-excursion estimate. Fix a root of endpoint distance m . For $1 \leq R \leq 2$, its contribution is at most C_L/m , which is bounded by $C'_L m^{-1} R^{-\kappa_L}$ after increasing the constant.

Suppose $R > 2$. At the right corner, the largest-label root has minimum endpoint distance. Thus logarithmic distance at least $\log R$ requires a vertex of distance at least Rm , and hence greater than $Rm/2$. Apply the global weighted row bound along the open path to that vertex, and the ordinary row bound thereafter. Bounding the closing edge by C_L/m gives

$$\frac{C_L}{m} \left(\frac{m}{Rm/2} \right)^{\kappa_L} \leq \frac{C'_L}{m} R^{-\kappa_L}.$$

At the left corner, the root has maximum endpoint distance. The corresponding event requires a vertex of distance at most m/R , hence below $2m/R$. The left weighted row bound gives the same estimate. Summing over the possible vertex positions changes only the constant; the global bounds include paths through the middle or the opposite corner.

Finally,

$$\sum_{\substack{m \in \mathbb{N} \\ A \leq m \leq B}} \frac{1}{m} \leq 1 + \log(B/A).$$

Summing the per-root estimate proves (116). A common positive exponent and common constants may be chosen over the finitely many cycle lengths and active corners. $\square$

For an active corner put

$$K_R(a, b) = \frac{1}{b} \beta q_R (a/b)^\beta e^{-q_R (a/b)^\beta}, \quad K_L(a, b) = \frac{1}{b} \alpha q_L (a/b)^{-\alpha} e^{-q_L (a/b)^{-\alpha}}.$$

These are deterministic ideal kernels on endpoint distances. They are not asserted to be globally uniform pointwise approximations to an averaged insertion probability. They are used below only through Proposition 6.5 and its domination estimates. In logarithmic coordinates their increment densities are, respectively, p_{β, q_R} and the reflection of p_{α, q_L} . Denote the appropriate density by p_σ . Reflection does not change its return density at zero, and

$$b_{\sigma, \ell} = \frac{1}{\ell} p_\sigma^{*\ell}(0).$$

For integer cutoffs $A \leq B$, let $\lambda_{n, \sigma, \ell}(A, B)$ be the sum of $\prod_{j=1}^\ell K_\sigma(a_j, a_{j+1})$, $a_{\ell+1} = a_1$, over distinct vertices in $[A, B]$, divided by ℓ . If a root interval is specified, retain only cycles whose largest-label root has its distance in that interval. These exact ideal sums, including their boundary corrections, will be the means used in the factorial-moment comparison.

Lemma 6.8 (Ideal traces). *For fixed ℓ , as $A \rightarrow \infty$, uniformly for $B \geq A$,*

$$\lambda_{n, \sigma, \ell}(A, B) = b_{\sigma, \ell} \log(B/A) + O_\ell(1) + O_\ell((1 + \log(B/A))/A). \quad (117)$$

Let $A_n = \lceil \exp((\log n)^{1/4}) \rceil$ and $B_n = \lfloor n/A_n \rfloor$. For a fixed interval $I = (a, b]$ with $0 < a < b < 1$, the ideal sum restricted to vertices in $[A_n, B_n]$ and root distance in $(n^a, n^b]$ satisfies

$$\lambda_{n, \sigma, \ell, I} = b_{\sigma, \ell} |I| \log n + O_\ell((1 + \log n)^K A_n^{-\kappa}) \quad (118)$$

for some $\kappa > 0$ and finite K .

Proof. Write $\lambda_\sigma(A, B) = \lambda_{n, \sigma, \ell}(A, B)$ and $p = p_\sigma$. For $y_0 = y_\ell = 0$, put

$$W(\mathbf{y}) = \prod_{j=0}^{\ell-1} p(y_j - y_{j+1}), \quad R(\mathbf{y}) = \max_{0 \leq j < \ell} y_j - \min_{0 \leq j < \ell} y_j.$$

The explicit density satisfies $p(x) + |p'(x)| \leq C e^{-g|x|}$. Consequently $\int e^{cR} W < \infty$ for some $c > 0$, by bounding one factor and integrating the remaining increments. In logarithmic coordinates, the continuous trace over an interval of width T is

$$\frac{1}{\ell} \int_{\mathbb{R}^{\ell-1}} (T - R)_+ W = b_{\sigma, \ell} T - d_{\sigma, \ell} + O_\ell(e^{-cT}), \quad d_{\sigma, \ell} = \frac{1}{\ell} \int_{\mathbb{R}^{\ell-1}} RW.$$

Indeed, $\int W = p^{*\ell}(0)$, and $(T - R)_+ = T - R + (R - T)_+$; exponential integrability bounds the last term. For $\ell = 1$, the range and boundary constant are zero.

The identity

$$K_\sigma(u, v) = \frac{p(\log u - \log v)}{v}$$

allows comparison with logarithmic cells $[\log m, \log(m+1))$. Their widths are $m^{-1}(1 + O(A^{-1}))$. Telescoping the product and using the derivative envelope gives total error $O_\ell((1+T)/A)$, including the upper boundary adjustment, where $T = \log(B/A)$. Removing repeated vertices costs

$O_\ell(A^{-1})$: bounded kernel rows and the entry bound C/v bound each prescribed coincidence by $C_\ell \sum_{m=A}^B m^{-2}$. Thus, uniformly for $B \geq A$,

$$\lambda_\sigma(A, B) = b_{\sigma, \ell} \log(B/A) - d_{\sigma, \ell} + O_\ell\left(\frac{1 + \log(B/A)}{A} + e^{-c \log(B/A)}\right).$$

This implies (117).

For the spatial estimate, set $P = \lfloor n^a \rfloor$ and $Q = \lfloor n^b \rfloor$. For large n , $A_n \leq P < Q < B_n$. Since the canonical root has maximum left depth and minimum right depth, respectively, the exact identities are

$$\lambda_{n, \sigma, \ell, I} = \begin{cases} \lambda_L(A_n, Q) - \lambda_L(A_n, P), & \sigma = L, \\ \lambda_R(P + 1, B_n) - \lambda_R(Q + 1, B_n), & \sigma = R. \end{cases}$$

Apply the refined expansion to both terms. The constants $d_{\sigma, \ell}$ cancel. Every interval appearing here has logarithmic width at least $c_I \log n$, so its exponential remainder is $O(A_n^{-1})$, since $\log A_n = o(\log n)$. The lattice errors are $O_\ell((1 + \log n)/A_n)$, and rounding changes the difference of logarithmic widths from $(b - a) \log n$ by $O(n^{-a})$. Hence

$$\lambda_{n, \sigma, \ell, I} = b_{\sigma, \ell} (b - a) \log n + O_\ell\left(\frac{1 + \log n}{A_n}\right),$$

which proves (118) with $K = \kappa = 1$. □

6.4 Mixed factorial moments

We now compare actual core cycles with the ideal sums. All constants in this comparison may depend on the fixed factorial-moment order. This dependence is harmless; no estimate uniform in that order is needed.

Lemma 6.9 (Overlaps and nearby ranks). *Fix a finite number s of proposed cycles of lengths at most L , all with vertices in endpoint cores of distances $[A, B]$. Suppose their joint weight is bounded by a product of a matrix having bounded row sums and entries at most C/b for a core target of distance b . The sum of such weights over assignments in which the actual labels in two specified vertex slots differ by at most a fixed D is at most*

$$C_{s, L, D} A^{-1} (1 + \log(B/A))^{s-1}. \quad (119)$$

For slots in the same corner this is equivalent to their endpoint distances differing by at most D . This includes equal vertices. The same bound, with a changed constant, holds for the union over every pair of slots.

Proof. Put $H = 1 + \log(B/A)$, and write $d_\sigma(i)$ for the endpoint distance of label i . Restrict F to each corner core, obtaining F_σ . Bounded rows and the target bound imply, for every fixed $j \geq 1$,

$$(F_\sigma^j)(u, v) \leq \frac{C_j}{d_\sigma(v)}, \quad \text{Tr}(F_\sigma^j) \leq C_j H.$$

For two distinct slots in one cycle, split the cycle into the two positive-length paths between them. For slots in different cycles, root each cycle at its selected slot. In either case, summing the other vertices of the affected cycles gives at most

$$\frac{C_{s, L}}{d_\sigma(u) d_\tau(v)}.$$

Each label has at most $2D + 1$ labels within distance D . Since endpoint distance is injective on each core and $2/(xy) \leq x^{-2} + y^{-2}$,

$$\sum_{\substack{u \text{ in the } \sigma \text{ core, } v \text{ in the } \tau \text{ core} \\ |u-v| \leq D}} \frac{1}{d_\sigma(u)d_\tau(v)} \leq (2D + 1) \sum_{m=A}^B m^{-2} \leq \frac{2(2D + 1)}{A}.$$

This includes mixed corners without requiring their cores to be disjoint. The unaffected cycles contribute at most $C_{s,L}H^{s-1}$, since $H \geq 1$, proving the specified-pair bound. Summing over at most $(sL)^2$ slot pairs proves the union bound. For slots in the same corner, label differences equal endpoint-distance differences. The argument also permits different matrices on different cycles when their bounds are uniform. $\square$

Consider any fixed finite list of categories c , each specifying a corner, a length, and either its whole core or a prescribed root interval. For categories of the same corner and length require the root intervals to be disjoint. Let $Y_{n,c}$ count the actual cycles in that category whose every vertex lies in its core, and let $\lambda_{n,c}$ be the exact ideal sum with precisely these restrictions.

Proposition 6.10 (Core factorial moments). *With A_n, B_n as in Lemma 6.8, for every fixed multi-index $\mathbf{r} = (r_c)$ there are $\kappa_{\mathbf{r}} > 0$ and $K_{\mathbf{r}} < \infty$ such that*

$$\left| \mathbb{E} \prod_c (Y_{n,c})_{r_c} - \prod_c \lambda_{n,c}^{r_c} \right| \leq C_{\mathbf{r}} (1 + \log n)^{K_{\mathbf{r}}} A_n^{-\kappa_{\mathbf{r}}}. \quad (120)$$

Here $(x)_r = x(x-1) \cdots (x-r+1)$.

Proof. Let $s = \sum_c r_c$, $m = \sum_c r_c \ell_c$, and $\mathcal{D} = \prod_c \ell_c^{r_c}$. The case $s = 0$ is immediate. Write $A = A_n$, $B = B_n$, and $H = 1 + \log(B/A)$.

Expand the mixed falling factorial into ordered lists of cycles and represent each cycle by an ordered vertex tuple. For large n , the endpoint cores are disjoint; together with category disjointness, this ensures that all selected permutation cycles are distinct and hence vertex-disjoint. Thus the actual moment is the sum of joint rank probabilities over globally distinct assignments, divided by $\mathcal{D}$. The product of exact ideal means has the same representation with ideal edge weights and distinctness required only within each cycle. Both sums retain the prescribed root restrictions.

First discard assignments with two labels within distance $2m$. Lemma 6.9, applied to the actual domination matrix and to the ideal kernels, bounds their contribution by $CA^{-1}H^{s-1}$.

Next discard nonmoderate edges. In endpoint coordinates, either matrix F has bounded rows, entries at most C/b , and, for a suitable $t > 0$,

$$\sum_b F(a, b) V_\sigma(b) \leq C V_\sigma(a), \quad V_L(a) = a^{-t}, \quad V_R(a) = a^t.$$

For any specified edge $a_i \rightarrow a_{i+1}$, cut that edge, bound it by C/a_{i+1} , and propagate the weighted-row estimate along the remaining path. This gives

$$\sum_{a_1, \dots, a_\ell \in [A, B]} \left(\prod_{j=1}^{\ell} F(a_j, a_{j+1}) \right) \frac{V_\sigma(a_i)}{V_\sigma(a_{i+1})} \leq C_\ell H, \quad a_{\ell+1} = a_1.$$

Writing $\gamma_L = \alpha$ and $\gamma_R = \beta$, the displayed ratio equals $x_\sigma(a_i, a_{i+1})^{t/\gamma_\sigma}$. Failure of moderation therefore makes it at least A^η for some $\eta > 0$. Summing over edges and the remaining cycles bounds all nonmoderate assignments by $CA^{-\eta}H^s$. This applies to the full domination matrix, including its exceptional terms, and to the ideal kernels.

On the remaining assignments, apply Proposition 6.5 jointly to all m marked vertices, including both corners in the same race. Its error is at most

$$C(A^{-\kappa} + (B/n)^\kappa) \prod_e H_{\sigma_e}(a_e, h_e).$$

The envelope kernels have bounded rows and target bounds, so their product sums to at most CH^s . Since $B/n \leq A^{-1}$, this contribution is $CA^{-\kappa}H^s$. Category restrictions remain in the comparison and are dropped only when bounding nonnegative error sums.

Combining the three contributions, reducing the positive exponent if necessary, and using $H \leq 1 + \log n$ gives

$$C_{\mathbf{r}}(1 + \log n)^{s+1} A_n^{-\kappa_{\mathbf{r}}}.$$

Division by $\mathcal{D} \geq 1$ proves (120), with $K_{\mathbf{r}} = s + 1$. $\square$

6.5 Power-law CLTs and localization

Proof of Theorem 1.6. Let $Y_{n,\sigma,\ell}$ denote the core counts and $\lambda_{n,\sigma,\ell}$ their exact ideal means. By Lemma 6.8,

$$\lambda_{n,\sigma,\ell} = b_{\sigma,\ell} \log n + O(1 + \log A_n).$$

Proposition 6.10 determines their limiting centered mixed moments: conversion from falling factorials to ordinary powers introduces only fixed powers of $\log n$, while

$$(1 + \log n)^K A_n^{-\kappa} \longrightarrow 0 \quad (K < \infty, \kappa > 0).$$

Thus the mixed moments of the counts centered at their exact ideal means and divided by their square roots agree, up to $o(1)$, with the corresponding products of normalized Poisson moments.

For completeness, this comparison can be made algebraically. Define the linear functional $\mathcal{L}_\lambda$ on polynomials by $\mathcal{L}_\lambda((x)_r) = \lambda^r$. Its centered moments $M_r(\lambda) = \mathcal{L}_\lambda((x - \lambda)^r)$ satisfy

$$M_0 = 1, \quad M_1 = 0, \quad M_{r+1}(\lambda) = \lambda \sum_{j=0}^{r-1} \binom{r}{j} M_j(\lambda).$$

After normalization by $\lambda^{(r+1)/2}$, induction gives the standard Gaussian moments as $\lambda \rightarrow \infty$. Hence the normalized core counts have limiting mixed moments equal to those of independent standard normals. Taylor expansion of the characteristic function of each linear combination, with the remainder controlled by even moments, gives convergence of characteristic functions by taking $n \rightarrow \infty$ first and then increasing the Taylor order. Lévy's theorem yields joint Gaussian convergence.

Since $\log A_n = O((\log n)^{1/4})$, replacing the ideal means by $b_{\sigma,\ell} \log n$ preserves this limit. Lemma 6.7 also gives

$$\mathbb{E} \left| C_{n,\ell} - \sum_{\sigma \text{ active}} Y_{n,\sigma,\ell} \right| = O(1 + \log A_n) = o(\sqrt{\log n}),$$

by summing the bounds for roots outside the core, cycles leaving the core, and the middle and inactive endpoint regions. The normalized error therefore vanishes in L^1 .

For each length, the corner limits combine with weights $\sqrt{b_{\sigma,\ell}}/B_\ell$, whose squares sum to one. This proves the asserted independent standard normal limits. The first factorial-moment comparison and the same omission bound give $\mathbb{E}C_{n,\ell} \sim B_\ell \log n$. $\square$

Proof of Theorem 1.7. Apply Proposition 6.10 to the categories (σ, ℓ, I_j) , retaining the largest-label root. Lemma 6.8 gives their exact ideal means as

$$\lambda_{n,\sigma,\ell,I_j} = b_{\sigma,\ell}|I_j| \log n + o(\sqrt{\log n}).$$

The mixed-moment and characteristic-function argument in the preceding proof therefore gives independent Gaussian limits for all normalized core counts.

Since $\overline{I_j} \subset (0, 1)$, every root in n^{I_j} lies in $[A_n, B_n]$ for sufficiently large n . Lemma 6.7 bounds the expected number of such cycles having a vertex outside the core by $O(1)$. Consequently the normalized difference between the full and core counts tends to zero in L^1 , proving the joint CLT. The first factorial-moment comparison and this same bound prove the stated mean asymptotics.

Finally, the largest-label root has minimum right distance and maximum left distance. If a vertex's logarithmic distance differs from the root's by more than $\delta \log n$, the corresponding distance ratio exceeds n^δ . The excursion bound in Lemma 6.7 therefore bounds the expected number of such cycles by

$$C_L(1 + \log n)n^{-\kappa_L \delta} \longrightarrow 0,$$

which proves localization for every vertex. $\square$

6.6 The critical pole

We first record the martingale criterion used for fixed points.

Proposition 6.11. *Let $I_{n,k} \in \{0, 1\}$ be adapted to a filtration, and put $p_{n,k} = \mathbb{E}(I_{n,k} \mid \mathcal{F}_{n,k-1})$. If $v_n \rightarrow \infty$ and*

$$\sum_k p_{n,k} = v_n + o_{\mathbb{P}}(\sqrt{v_n}), \quad \sum_k p_{n,k}^2 = o_{\mathbb{P}}(v_n),$$

then $(\sum_k I_{n,k} - v_n)/\sqrt{v_n} \Rightarrow \mathcal{N}(0, 1)$. If additionally $\mathbb{E} \sum_k p_{n,k} \sim v_n$, then $\mathbb{E} \sum_k I_{n,k} \sim v_n$.

Proof. Write $A_n = \sum_k p_{n,k}$. Stop each row before its compensator would exceed $2v_n$. The stopped and original rows agree on $\{A_n \leq 2v_n\}$, whose probability tends to one. Thus the stopped row inherits both convergence assumptions, and its normalized square sum is bounded by 2, so

$$\mathbb{E} \left[\frac{1}{v_n} \sum_k p_{n,k}^2 \right] \longrightarrow 0.$$

Hereafter work with the stopped row and suppress n .

Fix $t \in \mathbb{R}$, put $u = t/\sqrt{v}$, $a = e^{iu} - 1$, and define

$$S_k = \sum_{j \leq k} I_j, \quad A_k = \sum_{j \leq k} p_j, \quad Z_k = e^{iuS_k - aA_k}.$$

The conditional Bernoulli identity gives

$$\mathbb{E}(Z_k \mid \mathcal{F}_{k-1}) = Z_{k-1} e^{-ap_k} (1 + ap_k).$$

Since $A_k \leq 2v$, we have $|Z_k| \leq C_t$. Also $|e^{-ap}(1+ap) - 1| \leq C|a|^2 p^2$ for large n . Telescoping expectations therefore yields

$$|\mathbb{E}Z_N - 1| \leq C_t \mathbb{E} \left[\frac{1}{v} \sum_k p_k^2 \right] \rightarrow 0.$$

The first hypothesis implies $a(A_N - v) \rightarrow 0$ in probability. Both $e^{aA_N - iuv}$ and $e^{av - iuv}$ have modulus at most one, so their difference tends to zero in L^1 . Consequently

$$\mathbb{E}e^{iu(S_N - v)} = e^{av - iuv} \mathbb{E}Z_N + o(1) \rightarrow e^{-t^2/2},$$

because $v(e^{iu} - 1 - iu) \rightarrow -t^2/2$. Lévy's continuity theorem, followed by removal of stopping, proves the CLT. Finally, $\mathbb{E} \sum_k I_{n,k} = \mathbb{E} \sum_k p_{n,k}$, proving the expectation assertion. $\square$

Proof of Theorem 1.8. Write $c = c_L$. The profile assumptions give

$$\theta_{n,i} \asymp \frac{n}{i}, \quad \sum_{i=1}^n \left| \theta_{n,i} - \frac{cn}{i - 1/2} \right| \leq Cn. \quad (121)$$

Indeed, the error from the pole is bounded by an integrable power near zero and is bounded on intervals separated from zero. Comparing with the pure-pole integrals and their Riemann sums therefore gives

$$G_n(t) = ct \log(1/t) + O(t + n^{-1}), \quad D_n(t) = c \log(1/t) + O(1 + (nt)^{-1}), \quad (122)$$

where $G_n(t) = n^{-1} \sum_i (1 - e^{-t\theta_{n,i}})$ and $D_n(t) = n^{-1} \sum_i \theta_{n,i} e^{-t\theta_{n,i}}$. The same rate comparison shows that deleting at most m labels leaves total rate at least

$$n\{c \log(n/m) - C\}.$$

To see this, restrict the harmonic comparison to $i \geq m$: the deleted terms have total at most $m/m = 1$.

Choose a sufficiently small fixed $\varepsilon > 0$, and put

$$A = \lceil \sqrt{n} \rceil, \quad B = \lfloor \varepsilon n \rfloor, \quad z_m = \log(n/m).$$

Let W_m be the rate remaining before draw m , and J_m indicate that label m remains. Then $p_{n,m} = \theta_{n,m} J_m / W_m$.

For $A \leq m \leq B$, test the arrival count at

$$t_- = \frac{m}{16cnz_m}, \quad t_+ = \frac{8m}{cnz_m}.$$

The population estimates place its means below $m/2$ and above $2m$, respectively. Chernoff bounds, the remaining-rate lower bound, and the estimate for $D_n(t_-)$ give

$$\begin{aligned} W_m &\geq n(cz_m - C), \\ \mathbb{E}W_m &\leq cnz_m + Cn(1 + \log z_m + z_m/m), \\ \mathbb{P}(J_m = 0) &\leq C(z_m^{-1} + m^{-1}). \end{aligned}$$

For the second estimate, bound W_m by the surviving rate at t_- when fewer than m arrivals have occurred. The complementary contribution is at most $Cn^2e^{-c'm} \leq Cn/m$, since $n \leq m^2$. For the third, use $\mathbb{P}(E_{n,m} \leq t_+) \leq \theta_{n,m}t_+$ and the lower arrival-tail bound.

For small enough ε , the first estimate gives $W_m \geq cnz_m/2$. Combining the three estimates with the sampled endpoint expansion yields

$$\mathbb{E} \left| p_{n,m} - \frac{1}{mz_m} \right| \leq \frac{C}{mz_m} \left\{ \frac{1 + \log z_m}{z_m} + \frac{1}{m} + (m/n)^\eta \right\}.$$

Here the reciprocal error follows from

$$\mathbb{E}|W_m - cnz_m| \leq \mathbb{E}W_m - cnz_m + 2Cn.$$

Integral comparison shows that the displayed probability errors have bounded sum.

Define $q_{n,m} = 1/(mz_m)$ on $[A, B]$ and zero elsewhere. Before A , the deterministic bound $p_{n,m} \leq C/(mz_m)$ has bounded sum. Beyond B , the rates have uniform positive lower and upper bounds, so Lemma 6.6 gives $\sum_{m>B} \mathbb{E}p_{n,m} = O(1)$. Thus

$$\sum_m \mathbb{E}|p_{n,m} - q_{n,m}| = O(1), \quad \sum_m q_{n,m} = \log \log n + O(1), \quad \sum_m q_{n,m}^2 = O(1).$$

Since $0 \leq p_{n,m}, q_{n,m} \leq 1$, these imply

$$\sum_m p_{n,m} = \log \log n + O_{\mathbb{P}}(1), \quad \mathbb{E} \sum_m p_{n,m} = \log \log n + O(1), \quad \mathbb{E} \sum_m p_{n,m}^2 = O(1).$$

Proposition 6.11 proves the fixed-point CLT and mean asymptotic.

Now fix $\ell \geq 2$. Choose a sufficiently large fixed $M = M_\ell$, and decrease ε if necessary. Cycles meeting labels below M number at most M ; cycles meeting labels above B have bounded expected number by Lemma 6.6. It remains to consider cycles contained in $[M, B]$.

Put $L_j = \log(n/j)$. The deleted-rate lower bound and the arrival estimate at a small multiple of $j/(nL_j)$, combined with the exponential spacing moments, give an insertion domination kernel

$$F_n(i, j) = \frac{C_\ell}{iL_j} \exp\left(-\gamma_\ell \frac{j}{iL_j}\right), \quad M \leq i, j \leq B.$$

This estimate is uniform under the finitely many deletions and rank shifts needed for an ℓ -cycle. The arrival-tail error is absorbed into the displayed exponential because $j/(iL_j) \leq j$. Hölder's inequality therefore bounds each cycle probability by the product of its F_n -weights.

Root the cycle at its largest label $m = i_1$, and set

$$z = L_m, \quad v_j = \log(m/i_j) \geq 0, \quad S = \sum_{j=1}^{\ell} v_j, \quad v_1 = v_{\ell+1} = 0.$$

A closed path has some drop $v_j - v_{j+1} \geq S/(2\ell^2)$. Hence

$$\sum_{j=1}^{\ell} \frac{i_{j+1}}{i_j L_{i_{j+1}}} \geq \frac{e^{S/(2\ell^2)}}{z + S}.$$

The elementary bound $e^{-\gamma x} \leq Cx^{-1/2}$ then gives

$$\exp\left(-\gamma_\ell \frac{e^{S/(2\ell^2)}}{z+S}\right) \leq C_\ell z^{1/2} e^{-S/(8\ell^2)}.$$

Since every $L_{i_j} \geq z$ and $\ell \geq 2$, writing $a = 1/(8\ell^2)$ yields

$$\prod_{j=1}^{\ell} F_n(i_j, i_{j+1}) \leq \frac{C_\ell}{mz^{3/2}} \prod_{j=2}^{\ell} \frac{1}{i_j} \left(\frac{i_j}{m}\right)^a.$$

Each remaining vertex sums to at most $1 + 1/a$, even after dropping distinctness. The contribution rooted at m is therefore at most $C_\ell/(mL_m^{3/2})$. Finally,

$$\sum_{m=M}^B \frac{1}{mL_m^{3/2}} \leq C \left(1 + \int_{\log(1/\varepsilon)}^{\infty} z^{-3/2} dz\right) < \infty.$$

Adding the omitted cycles proves the uniform expectation bound; finitely many smaller n are absorbed into the constant. $\square$

6.7 The sampling grid and Sukhatme weights

The midpoint convention keeps the statement of the profile assumption simple. The following observation connects it to the exact weights in Corollary 1.9.

Lemma 6.12 (Interior-grid sampling). *The conclusions of Theorems 1.6 and 1.7 also hold if the sampling rule in Assumption 1.5 is replaced by*

$$\theta_{n,i} = f(i/(n+1)), \quad 1 \leq i \leq n.$$

The same replacement is valid in Theorem 1.8.

Proof. All sample points remain in $(0, 1)$. The endpoint distances are now $m/(n+1)$ at both ends. On an active endpoint block the deterministic expansions used above become

$$\theta_{n,n-m+1} = c_R(m/(n+1))^\beta (1 + O((m/n)^\eta)), \quad \theta_{n,m} = c_L(m/(n+1))^{-\alpha} (1 + O((m/n)^\eta)).$$

Compared with their leading expressions on the midpoint grid, these add relative errors $O(m^{-1} + n^{-1})$. In the proof of Lemma 6.1, the mesh is $1/(n+1)$ rather than $1/n$; division of the sums by n adds a factor $1 + 1/n$. The error in an unweighted sum remains $O(n^{-1})$, and that in a weighted sum remains $O((nt)^{-1})$. Thus the population and quantile estimates, including their leading constants, are unchanged. Endpoint comparability and the positive upper and lower bounds off the active neighborhoods are also unchanged.

The insertion and cycle arguments use precisely these deterministic estimates and bounds. The additional error on their core is $O(A_n^{-1} + n^{-1})$, which is already allowed by the mixed factorial moment estimates. The bounds outside the core remain uniform. This proves the two power-law assertions. For the critical pole the same sum-integral estimates give $G_n(t) = c_L t \log(1/t) + O(t + n^{-1})$ and $D_n(t) = c_L \log(1/t) + O(1 + (nt)^{-1})$ in the range used above; global comparability with n/i persists. Hence the critical fixed-point and longer-cycle arguments apply as well. This is a verification of the proof inputs for the second grid; it does not assert a coupling between the cycle counts on the two grids. $\square$

7 Independent-clock extensions

The bulk density (9) uses the exponential form of the Luce race, but the last-position estimate has a distribution-free core. This identifies a broader class of ranking models for which the same fixed-point strategy is available.

Let $T_1, \dots, T_n$ be independent continuous random variables with densities g_i and survival functions $\bar{G}_i$. Order them increasingly and write

$$R_i = 1 + \sum_{j \neq i} \mathbf{1}\{T_j < T_i\}, \quad S(t) := \sum_{i=1}^n \bar{G}_i(t).$$

As before, put $k_m = n - m + 1$.

Proposition 7.1. *Let $1 \leq M \leq n$, choose B with $B - 1 \geq 2M$, and let s satisfy $S(s) = B$. Suppose that an integrable function h satisfies*

$$g_{k_m}(t) \leq h(t), \quad 1 \leq m \leq M, \quad t \geq s. \quad (123)$$

Then, for a universal $c > 0$,

$$\sum_{m=1}^M \mathbb{P}(R_{k_m} = k_m) \leq M e^{-cB} + 2 \int_s^\infty h(t) dt. \quad (124)$$

Proof. Conditioning on $T_{k_m} = t$ gives the exact identity

$$\mathbb{P}(R_{k_m} = k_m) = \int_{-\infty}^\infty g_{k_m}(t) \mathbb{P}\left(\sum_{j \neq k_m} \mathbf{1}\{T_j > t\} = m - 1\right) dt.$$

For $t \leq s$, the survivor count in parentheses has mean at least $S(t) - 1 \geq B - 1 \geq 2M$, so its lower tail is at most e^{-cB} by a Chernoff bound. The integral over this range, summed over m , is at most $M e^{-cB}$.

For fixed t , couple all the inner probabilities using one independent sample of the clocks. If $N(t) = \sum_j \mathbf{1}\{T_j > t\}$, the event in the last display can hold only for $m = N(t)$ or $m = N(t) + 1$. Hence the sum of its indicators over $m \leq M$ is at most two. On $t \geq s$, use (123), take expectations, and integrate to obtain the second term in (124). $\square$

Suppose more generally that the clock attached to label i/n has a limiting distribution function $G(x, t)$ and density $g(x, t)$. If

$$F(t) := \int_0^1 G(x, t) dx, \quad d(t) := \int_0^1 g(x, t) dx,$$

assume that F is strictly increasing on the range under consideration and that $d(t_y) > 0$. With $t_y = F^{-1}(y)$, the natural bulk density is

$$\rho_T(x, y) = \frac{g(x, t_y)}{d(t_y)}. \quad (125)$$

Thus a bulk local-rank theorem together with Proposition 7.1 gives the same architecture as the fixed-point proof above, provided one can choose triangular-array parameters M_n, B_n, s_n, h_n for

which the right side of (124) tends to zero. A candidate bulk density is not enough by itself: one must still prove compensator convergence, or an equivalent microscopic local law.

Longer cycles require another input. Their late closing label need not be a terminal label, so Proposition 7.1 alone does not control them. In the exponential case, Lemma 5.3 and Propositions 5.4 and 5.5 overcome this obstruction by swapping a finite insertion path into the background, charging exceptional rates, and marking the edge into the largest cycle label. The finite-path lemma is distribution-free; the truncation and shell bounds for another clock family must come from suitable density comparisons and endpoint envelopes.

A concrete testing ground is the common-shape Gamma ranking family of [Ste90]. Shape one is the exponential race and hence the Luce model. For fixed Gamma shape and terminal rates bounded below, the terminal densities have a common polynomial-times-exponential envelope, so Proposition 7.1 applies; a bulk local law remains to be established. The independent-utility ranking models of [Thu27] are another natural target, provided their terminal densities admit an integrable envelope and the corresponding bulk local limit is available. Separately, the finite-dimensional criterion of [Muk16] already yields the cyclic trace law (18) for permuton-sampled permutations with strictly positive continuous densities, Mallows permutations with $n(1 - q_n)$ convergent, and certain continuous exponential-family models.

The extension also has sharp limits. For each fixed Ewens parameter $\vartheta > 0$, asymptotically uniform one-point marginals coexist with limiting ℓ -cycle mean ϑ/ℓ [Tav21]; thus a microscopic hypothesis remains essential. For fixed $q \in (0, 1)$, Mallows permutations lie in a localized regime in which bounded-cycle counts have linear means and joint Gaussian rather than Poisson fluctuations [HMV23]. These examples underline the central point: the macroscopic limit predicts the form of the cycle constants only after a model-specific argument has reached the microscopic scale.

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